

Indian Institute of Technology Hyderabad
Department of Electrical Engineering

CMOS Digital Design Project Report

Project Title: Digital Lock System

Course: CMOS Digital Design

Submitted by

Name: Nimmala Sree Prabhath
Roll No: EE25MTECH11010

Name: Joel Giby
Roll No: EE25MTECH11006

Under the Guidance of

Dr. Kapil Jainwal
Assistant Professor
Department of Electrical Engineering

25 November 2025

Contents

Abstract 1

1 Problem Statement 2

2 System Overview 3

2.1 Block Diagram Description . . . 3

3 Block-Level Design 4

3.1 4-bit Password Memory . . . 4

3.2 4-bit Comparator Design . . . 4

3.3 Unlock Logic . . . 5

4 Individual Block Schematics and Symbols 6

4.1 Inverter Gate . . . 6

4.1.1 Transistor-Level Schematic . . . 6

4.1.2 Symbol . . . 7

4.2 2-Input AND Gate . . . 8

4.2.1 Transistor-Level Schematic . . . 8

4.2.2 Symbol . . . 9

4.3 3-Input AND Gate . . . 10

4.3.1 Transistor-Level Schematic . . . 10

4.3.2 Symbol . . . 11

4.4 4-Input AND Gate . . . 12

4.4.1 Transistor-Level Schematic . . . 12

4.4.2 Symbol . . . 13

4.5 4-Input OR Gate . . . 14

4.5.1 Transistor-Level Schematic . . . 14

4.5.2 Symbol . . . 15

4.6 2-Input XNOR Gate . . . 16

4.6.1 Transistor-Level Schematic . . . 16

4.7 4-bit Comparator (Integrated) . . . 17

4.7.1 Transistor-Level Schematic . . . 17
4.7.2 Symbol . . . 22
4.7.3 Simulation . . . 23
4.8 C^2MOS Flip-Flop Cell . . . 24
4.8.1 Transistor-Level C^2MOS Flip-Flop Cell . . . 24
4.8.2 Symbol . . . 25
4.8.3 Simulation . . . 26
4.9 Johnson Counter . . . 27
4.9.1 Transistor-Level Johnson Counter Cell . . . 27
4.9.2 Symbol . . . 29
4.9.3 Simulation . . . 31
4.10 6T SRAM Cell . . . 33
4.10.1 Transistor-Level Memory Cell . . . 33
4.11 Sense Amplifier - OTA . . . 34
4.11.1 Schematic . . . 34
4.11.2 Symbol . . . 35
4.11.3 Simulation . . . 36
4.12 Integrated 6T SRAM + Sense Amplifier Block . . . 38
4.12.1 Combined Schematic . . . 38
4.12.2 Simulation . . . 39
4.13 Stored Passwords - A and B . . . 40
4.13.1 Passwords A Schematic . . . 40
4.13.2 Passwords A Symbol . . . 41
4.13.3 Passwords B Schematic . . . 42
4.13.4 Passwords B Symbol . . . 43

5 Complete Schematic **44**

6 Simulation and Results **49**
6.1 Simulation Setup . . . 49
6.2 Block Waveforms and Observations . . . 50

7 Conclusion **56**

References **57**

Abstract

This project presents the transistor-level design and simulation of a 4-bit Digital Lock System implemented in Cadence Virtuoso. The system stores two fixed 4-bit passwords using static memory structures and verifies a user-entered password through a custom comparison architecture designed at the transistor level. All functional blocks, including memory, comparison circuitry, and decision logic, are fully implemented using CMOS design principles. The overall operation of the lock is validated through transient simulations, confirming correct password detection and reliable system performance.

Chapter 1

Problem Statement

Design and simulate a transistor-level 4-bit Digital Lock System in Cadence Virtuoso with the following requirements:

- Two 4-bit passwords (Password A and Password B) must be stored statically using 6T SRAM cells.
- The system must accept a 4-bit user input password.
- The user-entered inputs are processed through two Johnson counters initialized with different starting states, ensuring pseudo-randomized internal sequencing before comparison.
- Two parallel comparator blocks must evaluate the bitwise matching between the processed user input and each stored password.
- The lock should open only if both stored passwords match the user input.
- All functional blocks, including SRAM storage, counters, comparison circuitry, and decision logic, must be implemented using transistor-level CMOS logic in Cadence Virtuoso.

Chapter 2

System Overview

2.1 Block Diagram Description

The high-level system consists of the following functional blocks:

- 4-bit Password Memory A implemented using 6T SRAM cells
- 4-bit Password Memory B implemented using 6T SRAM cells
- Two 4-bit Johnson Counter–Based Input Blocks, each initialized with different starting states to generate pseudo-randomized user input sequences
- Two 4-bit Comparator Blocks that match each Johnson-counter-generated input against the corresponding stored password
- Final Unlock Decision Block that produces the unlock signal only when both comparisons indicate a correct match

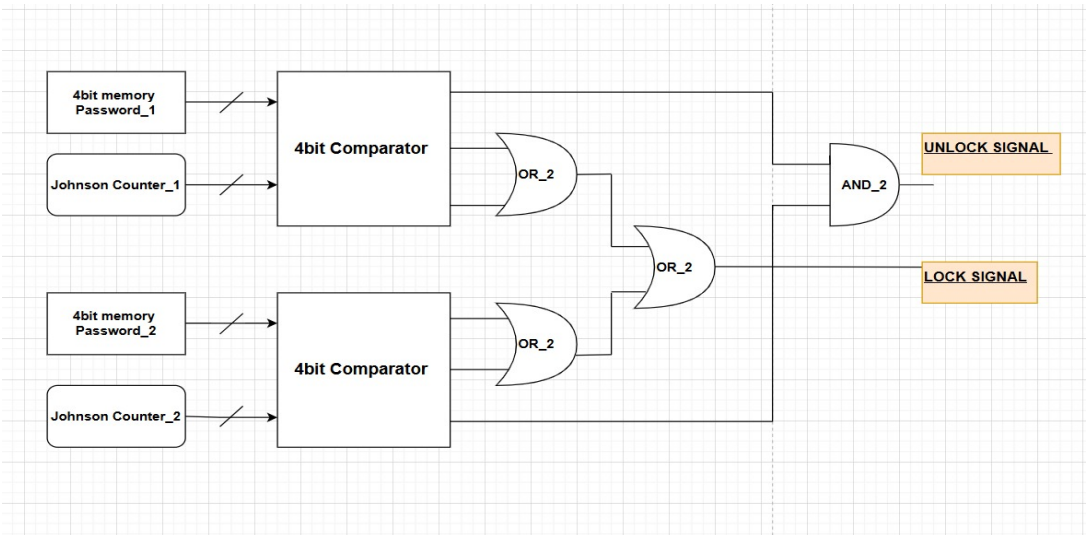


Figure 2.1: Overall Digital Lock System Block Diagram

Chapter 3

Block-Level Design

3.1 4-bit Password Memory

Each stored password uses four static memory cells. A 6T SRAM-style cross-coupled inverter cell is used to hold each bit (‘0‘ or ‘1‘). These cells maintain their state throughout the simulation.

3.2 4-bit Comparator Design

Each bit of the two 4-bit inputs $A = (A_3, A_2, A_1, A_0)$ and $B = (B_3, B_2, B_1, B_0)$ is first compared using a CMOS XNOR gate:

$$E_i = XNOR(A_i, B_i)$$

The 4-bit equality condition is computed as follows:

$$A = B = E_3 \cdot E_2 \cdot E_1 \cdot E_0$$

The bitwise greater signals are defined as:

$$G_i = A_i \cdot \overline{B_i}$$

Similarly, the bitwise less signals are defined as:

$$L_i = \overline{A_i} \cdot B_i$$

The complete 4-bit “greater than” logic is:

$$A > B = G_3 + (E_3 \cdot G_2) + (E_3 \cdot E_2 \cdot G_1) + (E_3 \cdot E_2 \cdot E_1 \cdot G_0)$$

The complete 4-bit “less than” logic is:

$$A < B = L_3 + (E_3 \cdot L_2) + (E_3 \cdot E_2 \cdot L_1) + (E_3 \cdot E_2 \cdot E_1 \cdot L_0)$$

3.3 Unlock Logic

The final unlock signal is generated as:

$$Lock_Open = Match_A \wedge Match_B$$

Implemented using a transistor-level AND gate.

Chapter 4

Individual Block Schematics and Symbols

4.1 Inverter Gate

4.1.1 Transistor-Level Schematic

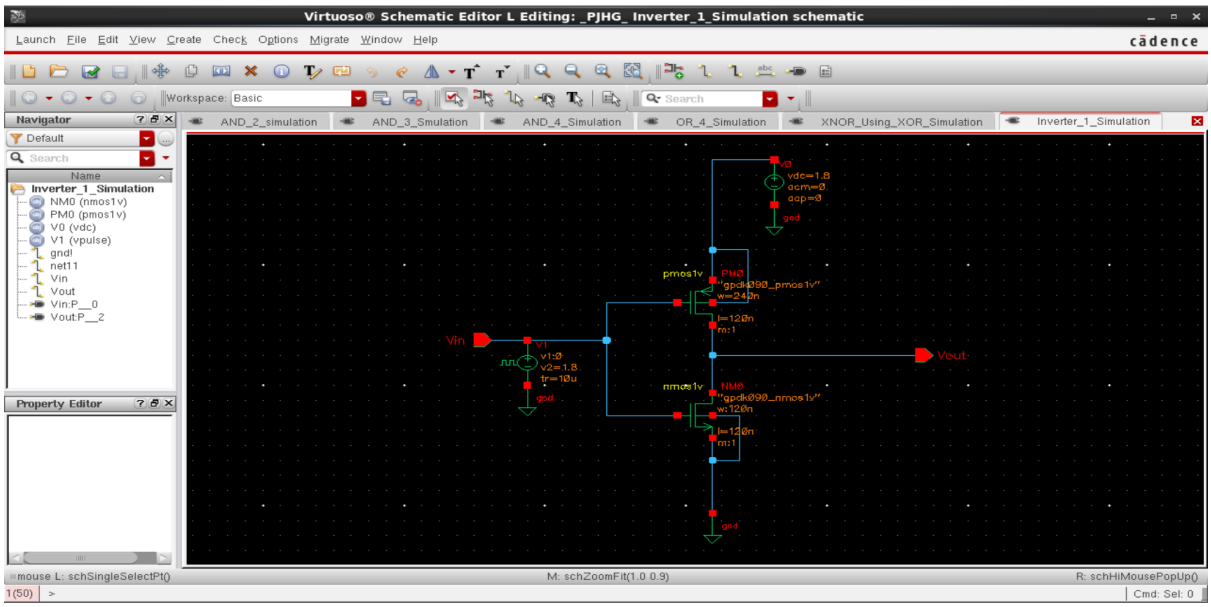


Figure 4.1: Inverter Schematic.

4.1.2 Symbol

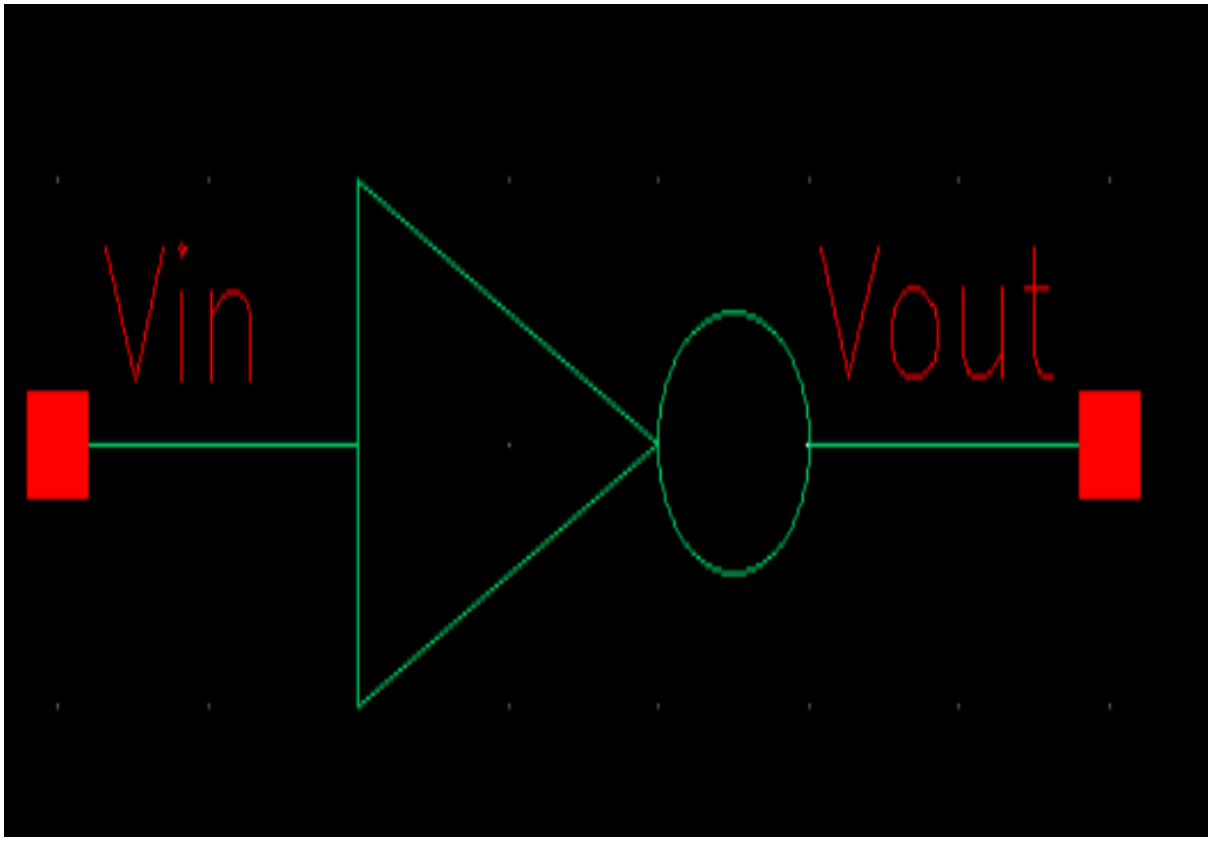


Figure 4.2: Inverter Symbol

4.2 2-Input AND Gate

4.2.1 Transistor-Level Schematic

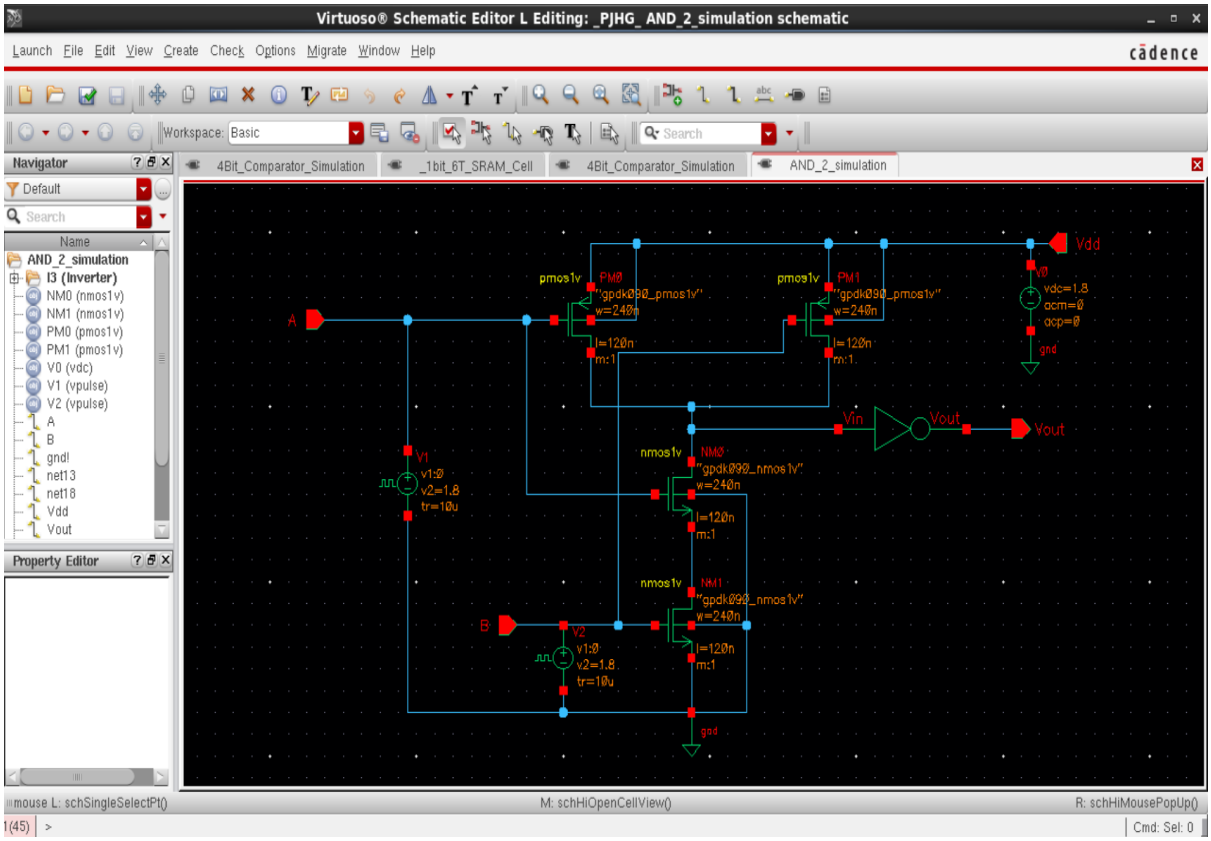


Figure 4.3: 2-input AND Gate Transistor-Level Schematic.

4.2.2 Symbol

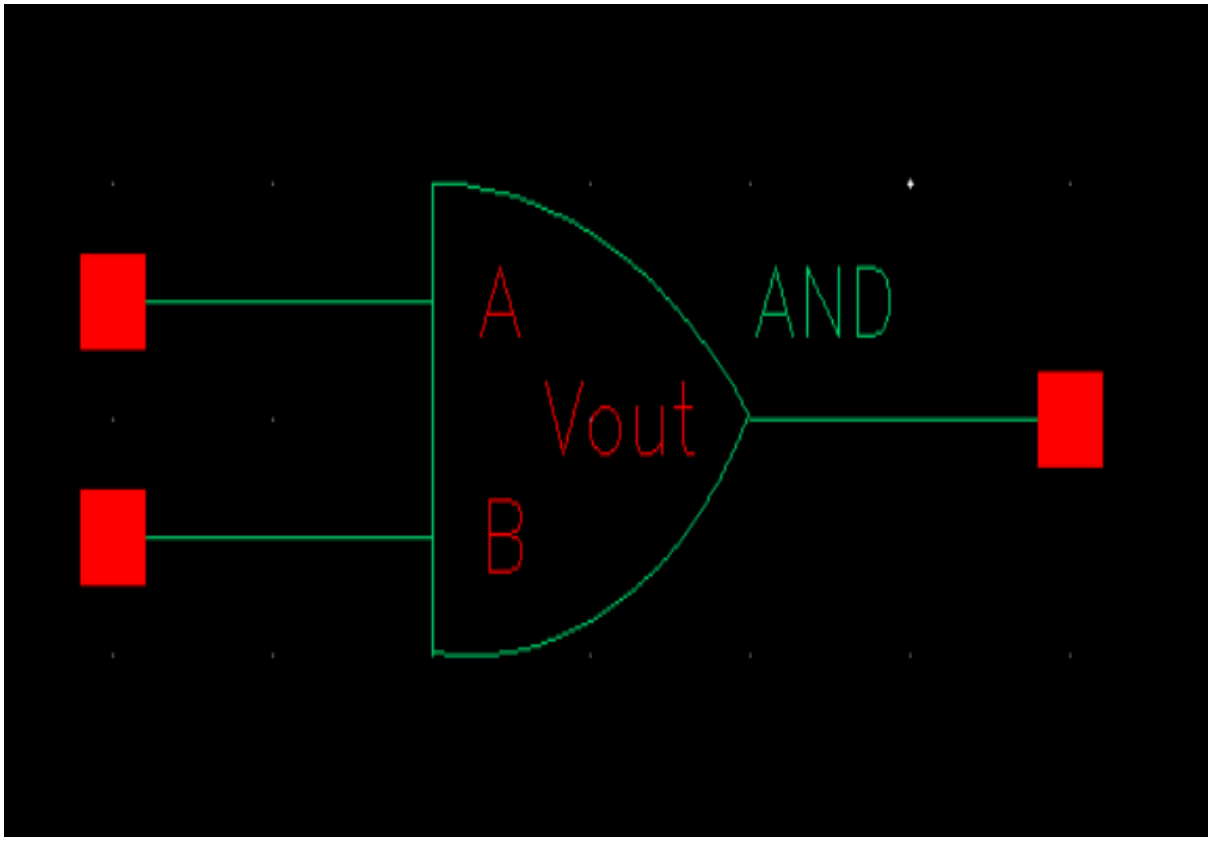


Figure 4.4: 2-input AND Gate Symbol

4.3 3-Input AND Gate

4.3.1 Transistor-Level Schematic

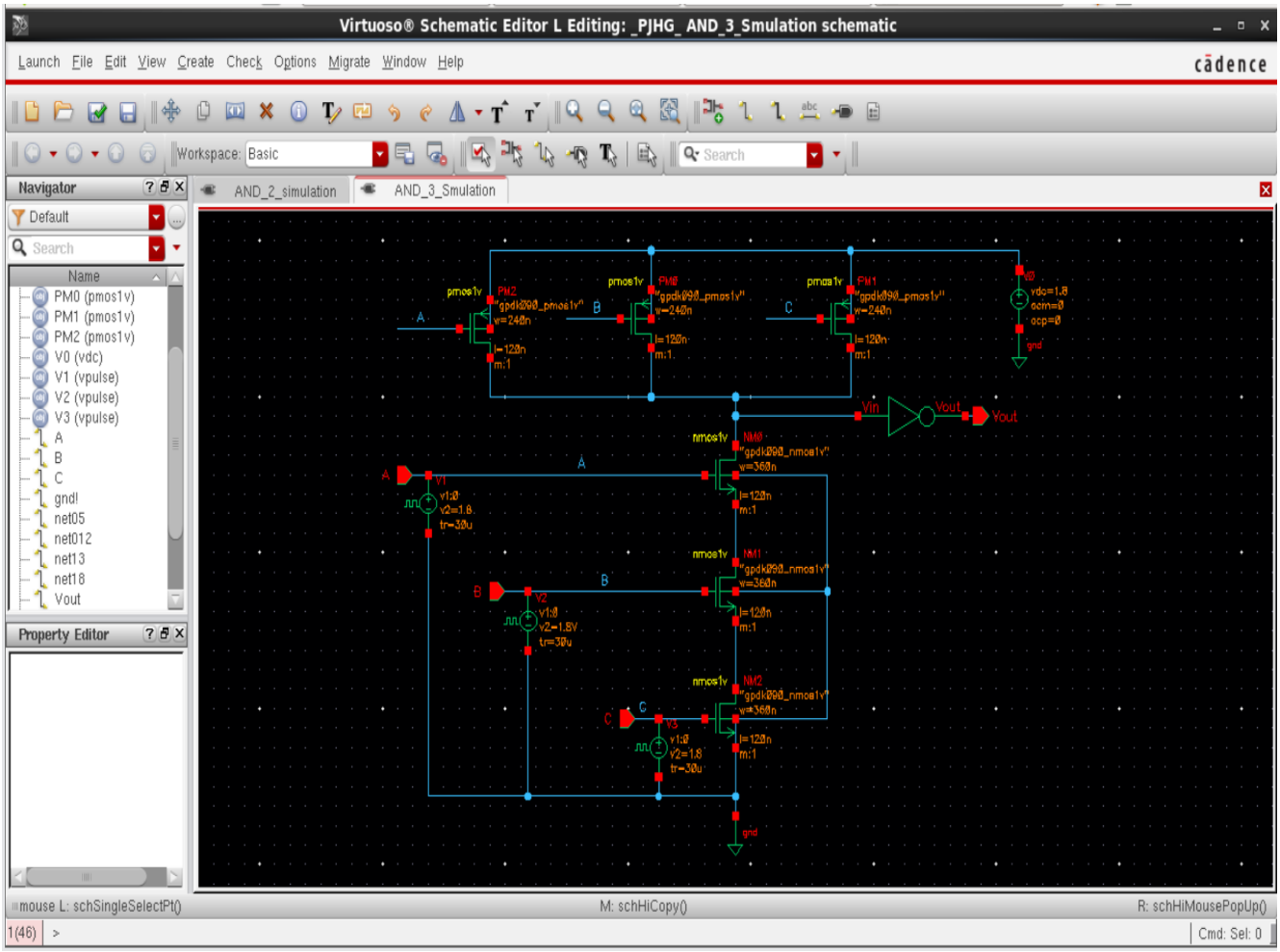


Figure 4.5: 3-input AND Gate Schematic.

4.3.2 Symbol

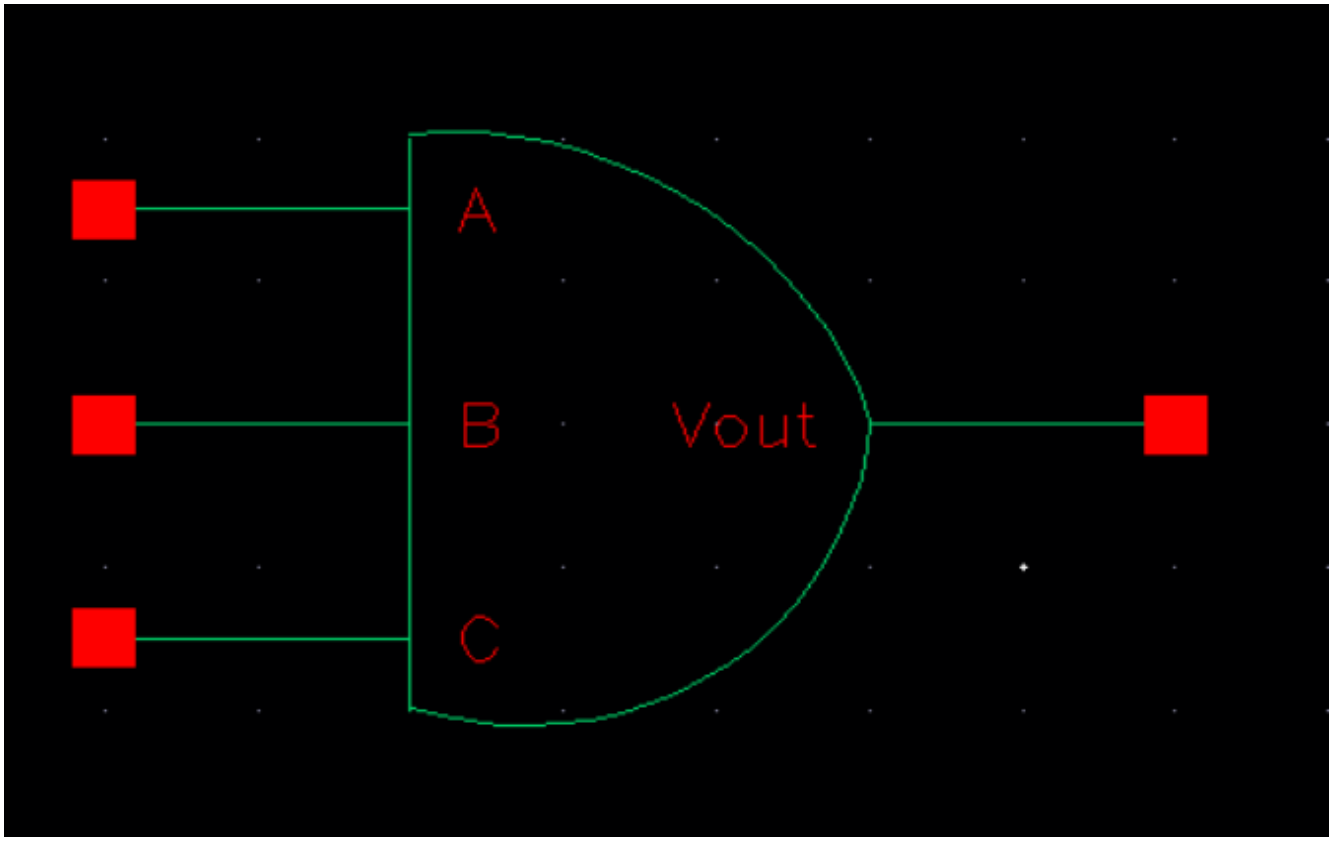


Figure 4.6: 3-input AND Gate Symbol.

4.4 4-Input AND Gate

4.4.1 Transistor-Level Schematic

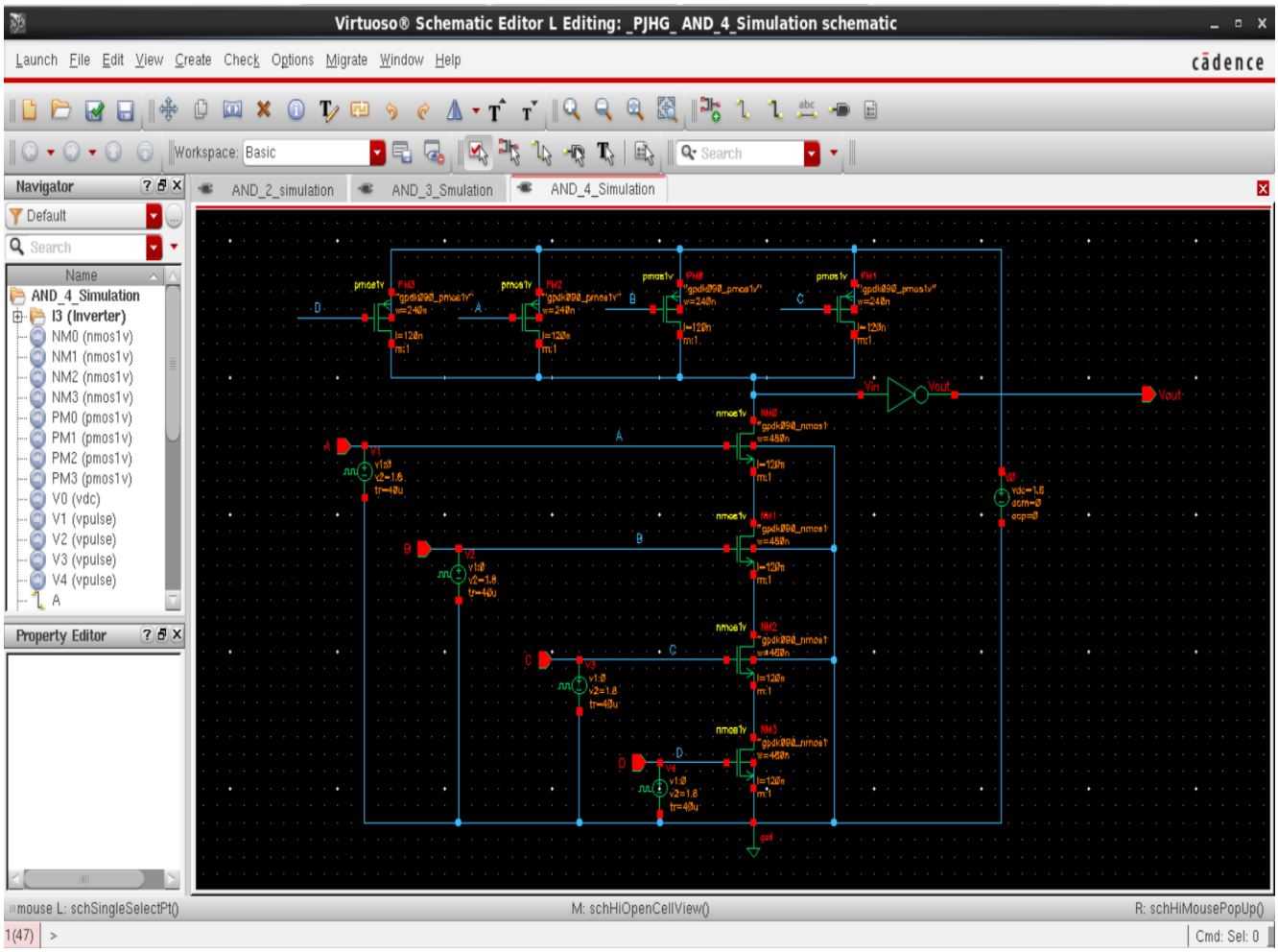


Figure 4.7: 4-input AND Gate Schematic.

4.4.2 Symbol

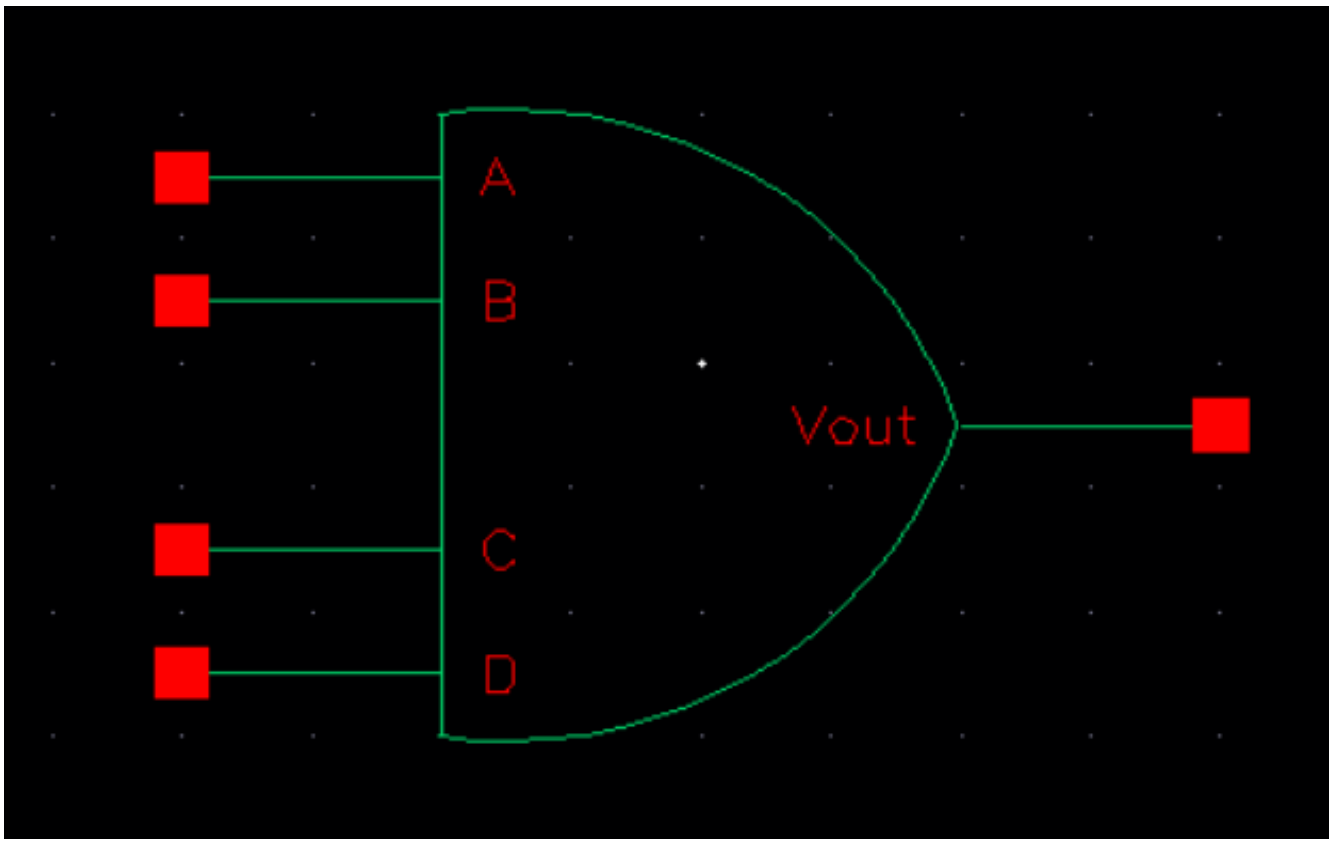


Figure 4.8: 4-input AND Gate Symbol.

4.5 4-Input OR Gate

4.5.1 Transistor-Level Schematic

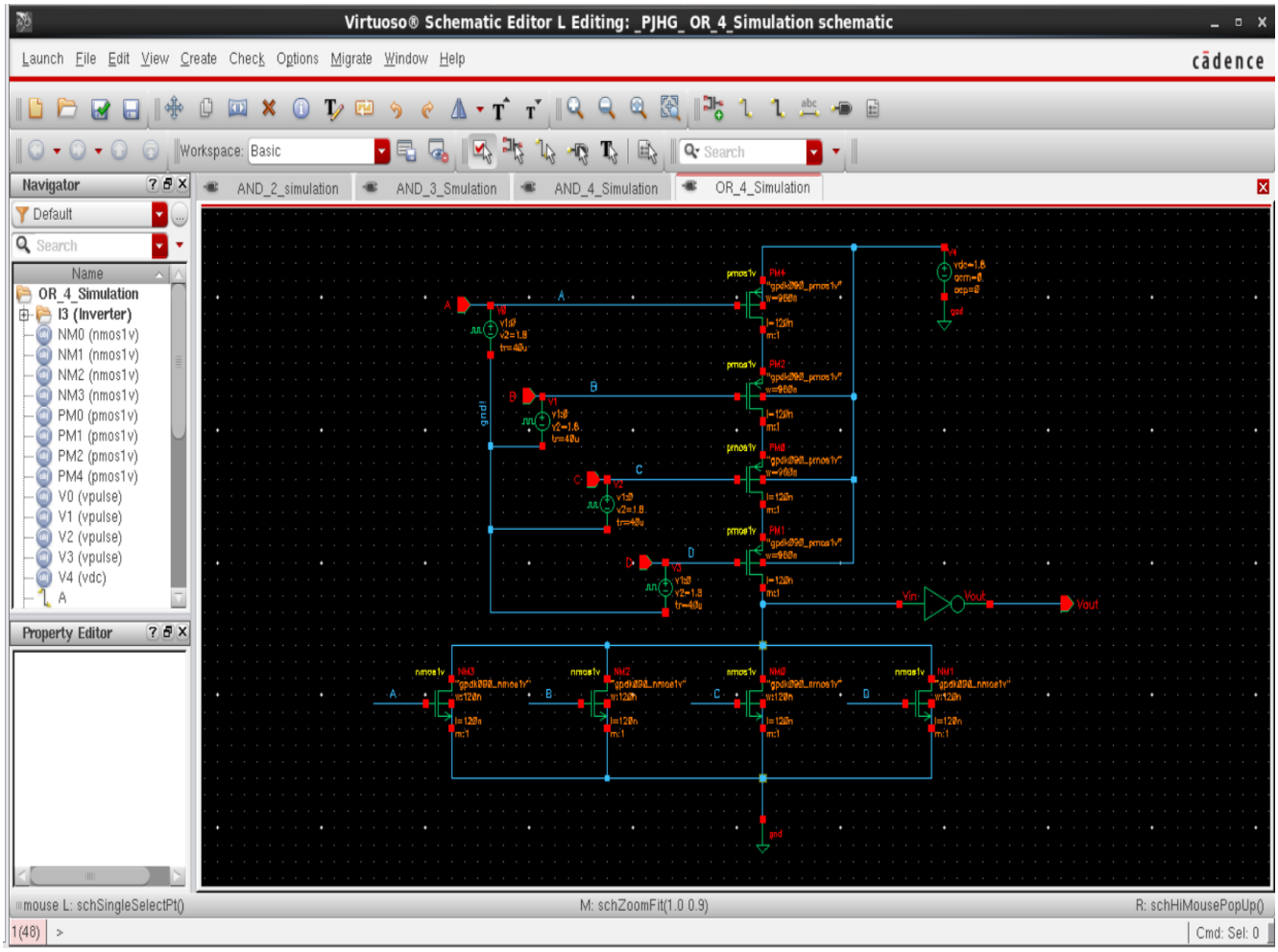


Figure 4.9: 4-input OR Gate Schematic.

4.5.2 Symbol

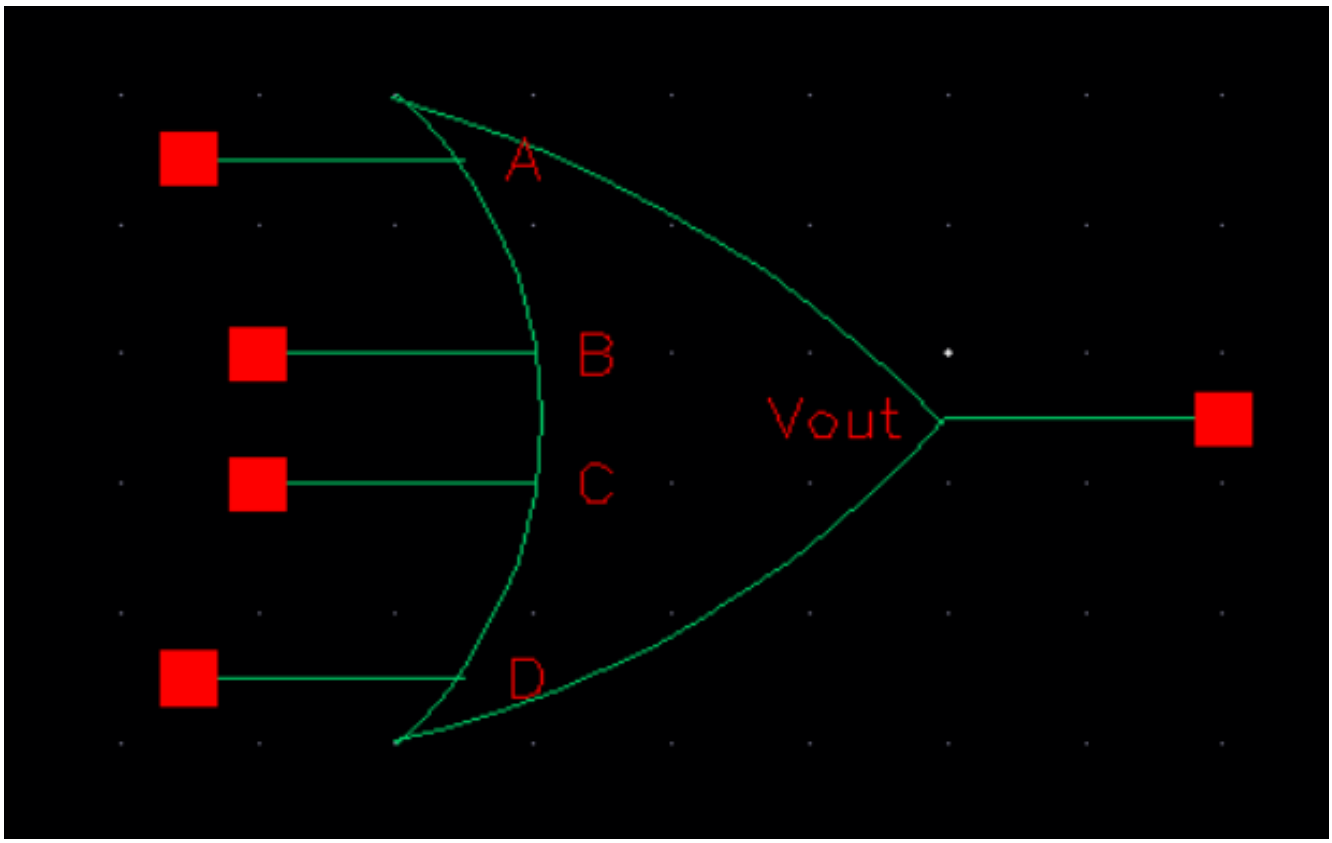


Figure 4.10: 4-input OR Gate Symbol.

4.6 2-Input XNOR Gate

4.6.1 Transistor-Level Schematic

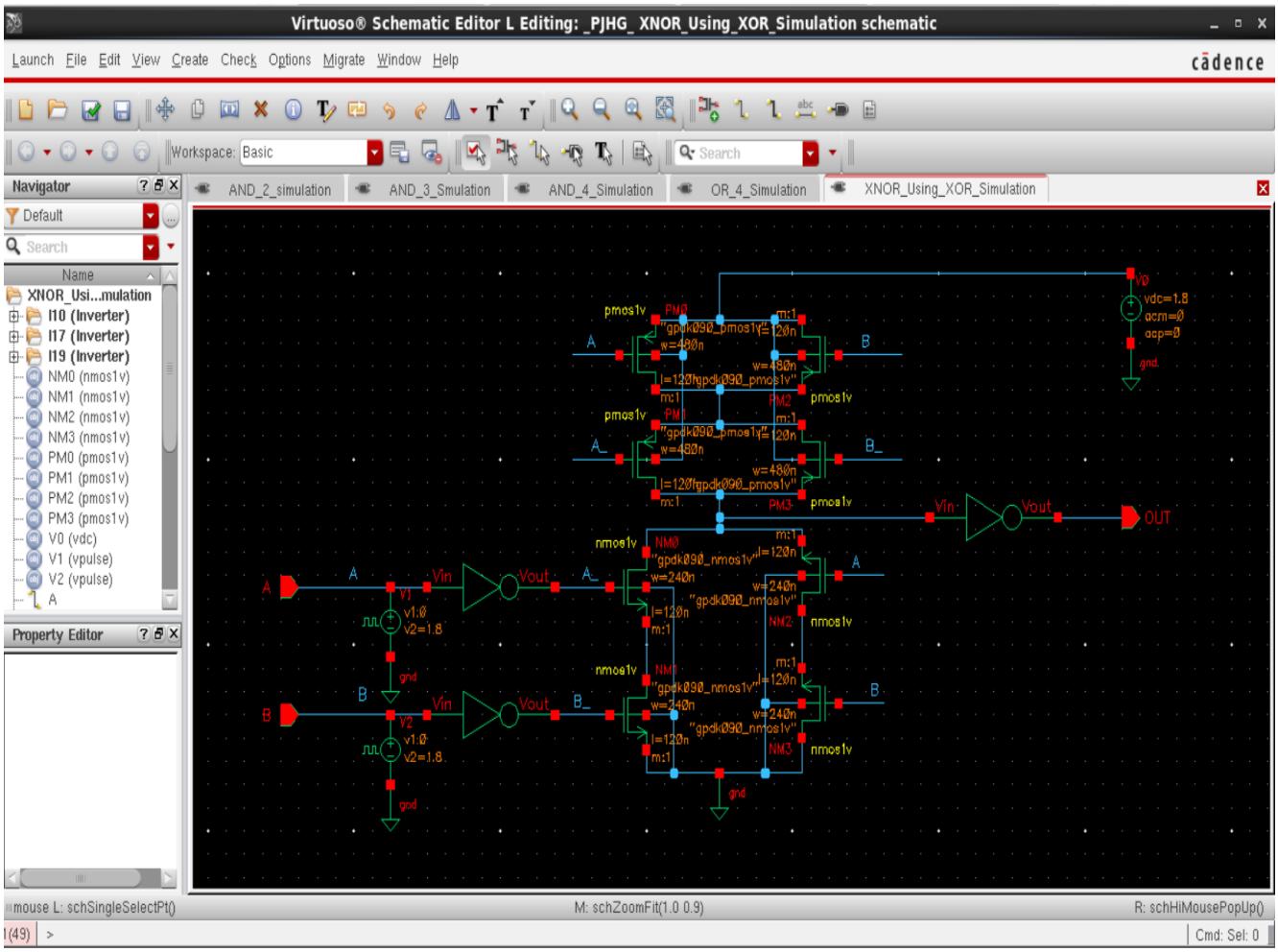


Figure 4.11: 2-input XNOR Transistor-Level Schematic.

4.7 4-bit Comparator (Integrated)

4.7.1 Transistor-Level Schematic

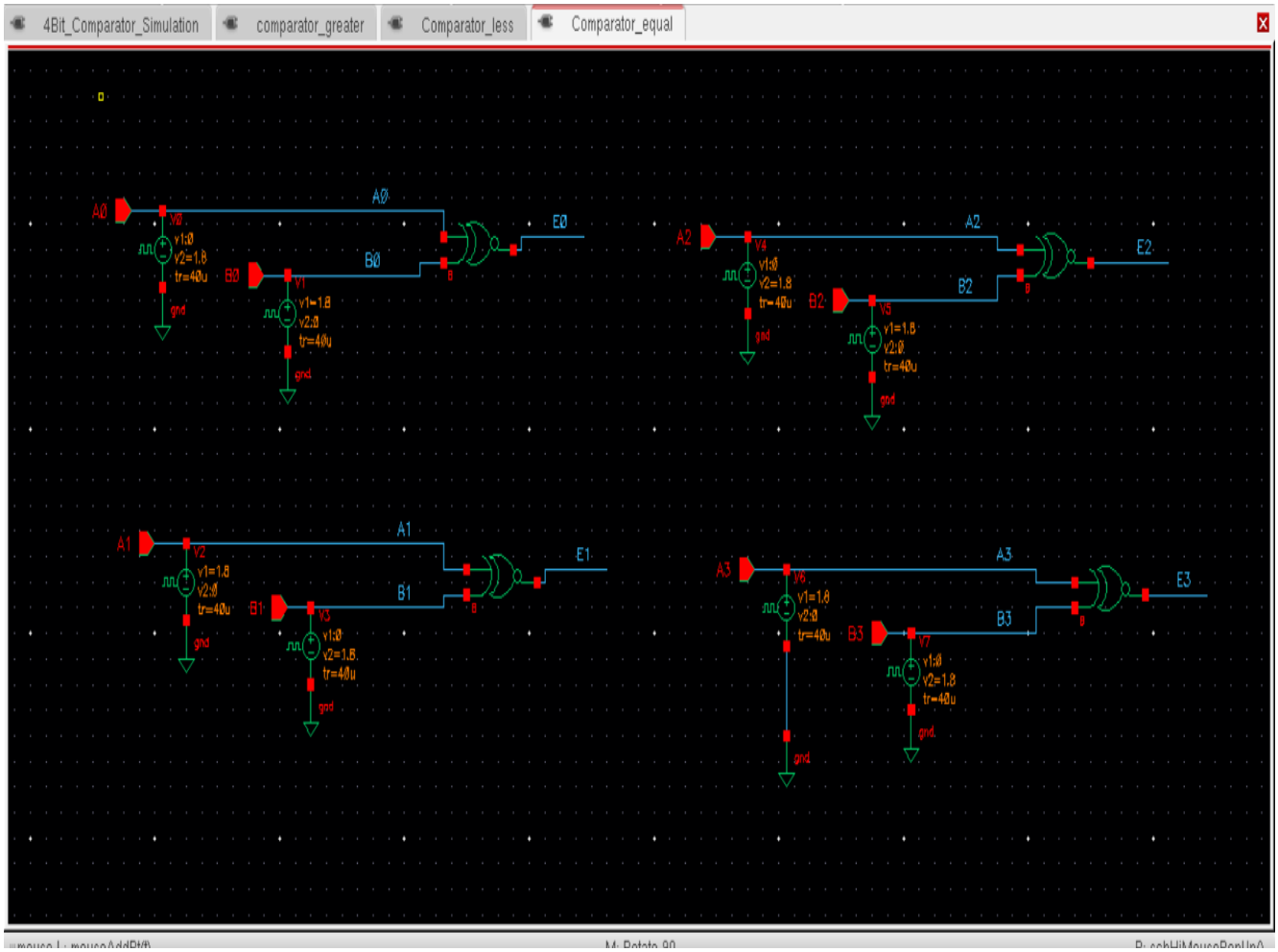


Figure 4.12: Schematic of the transistor-level 4-bit comparator implementing equality logic.

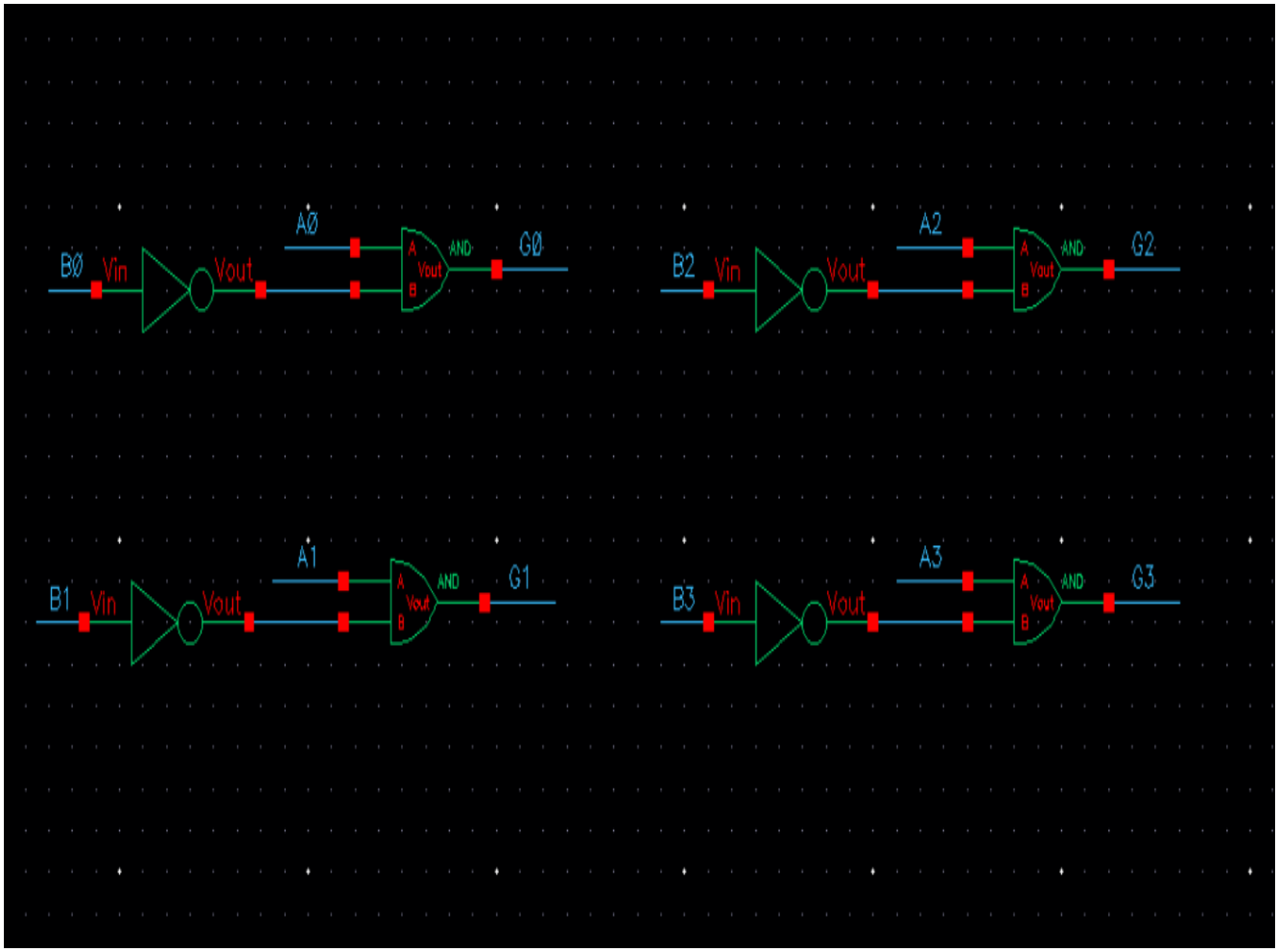


Figure 4.13: Schematic of the transistor-level 4-bit comparator implementing greater than logic.

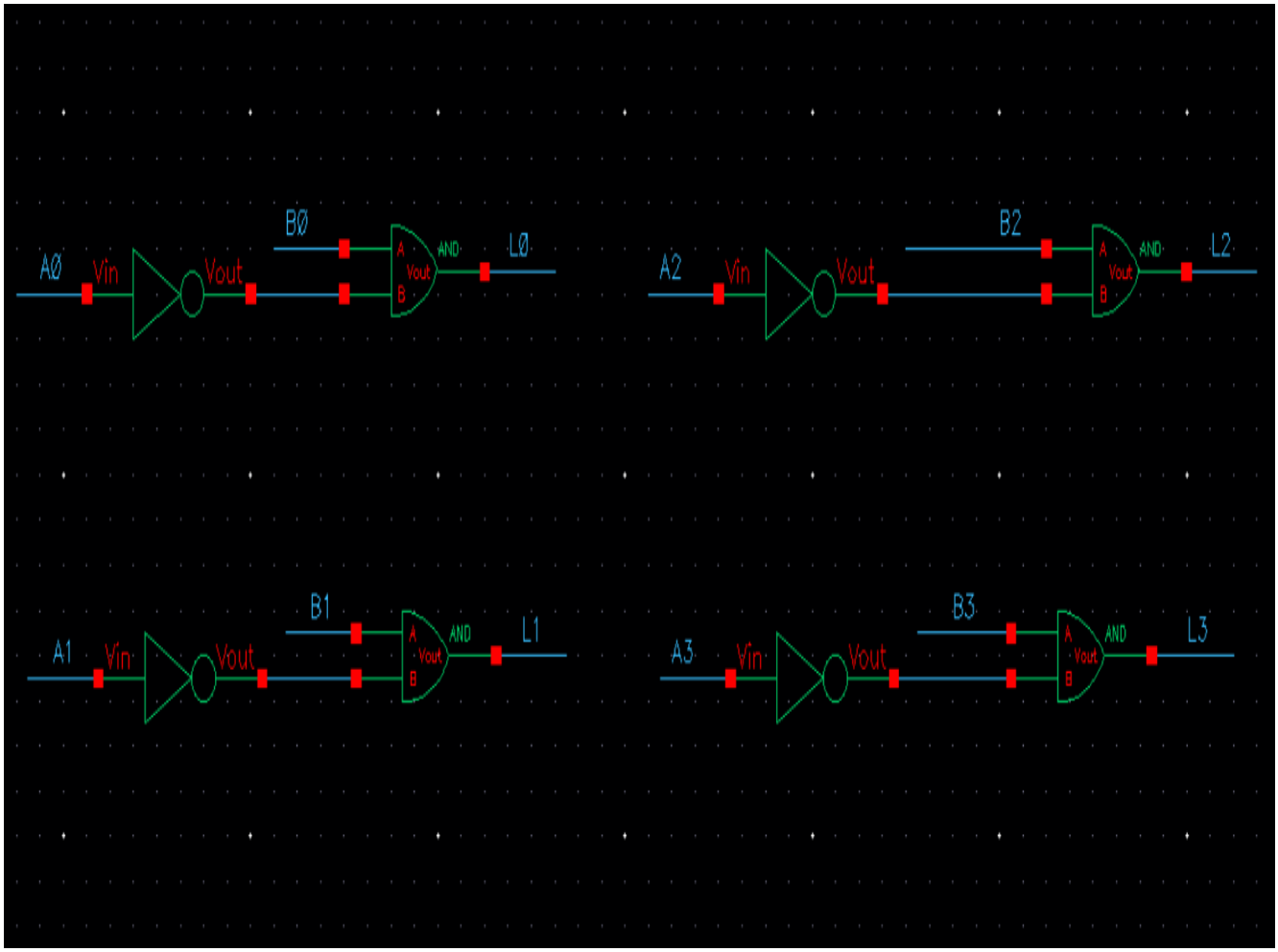


Figure 4.14: Schematic of the transistor-level 4-bit comparator implementing less than logic.

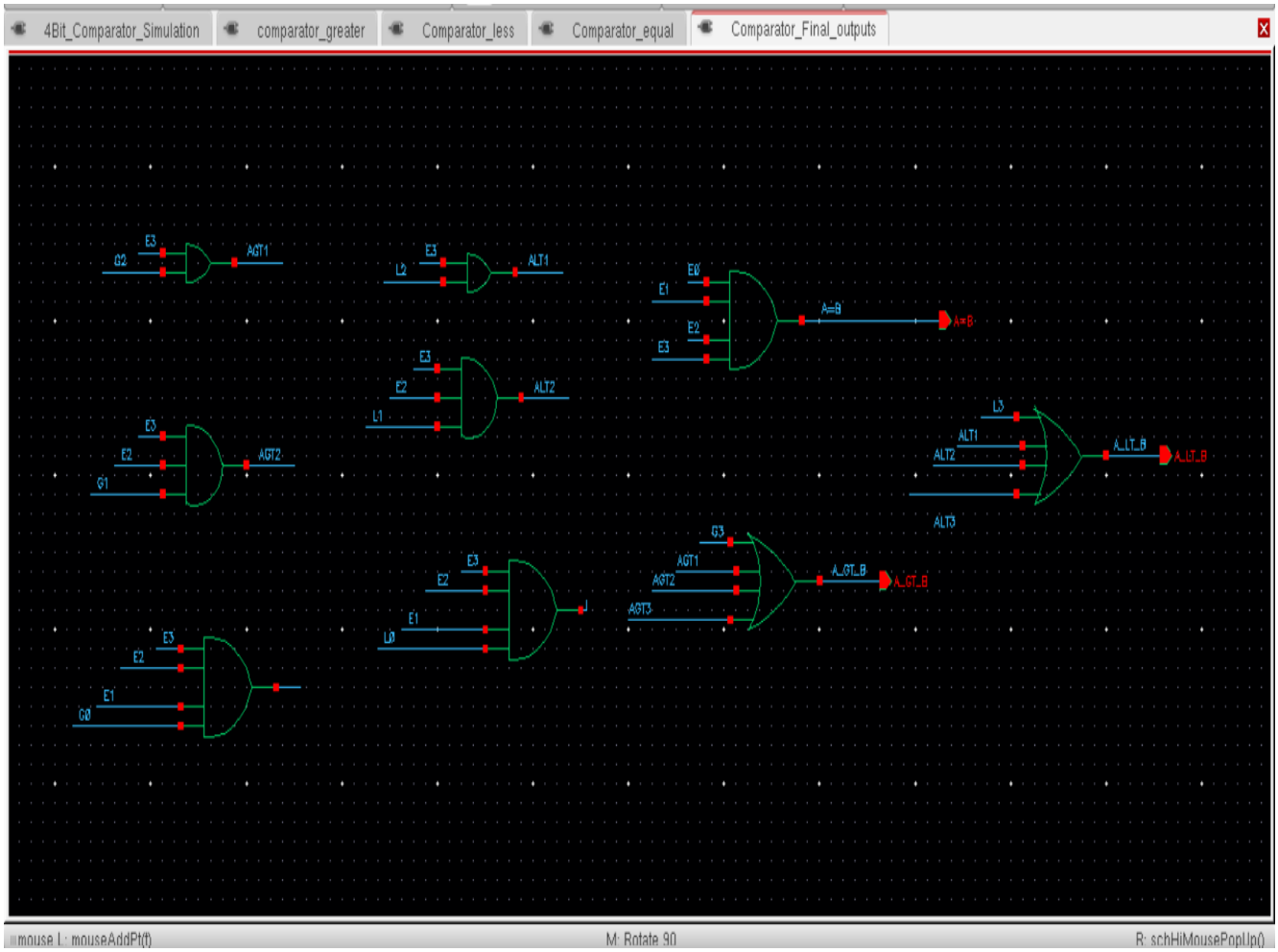


Figure 4.15: Transistor-level 4-bit comparator schematic with final outputs.

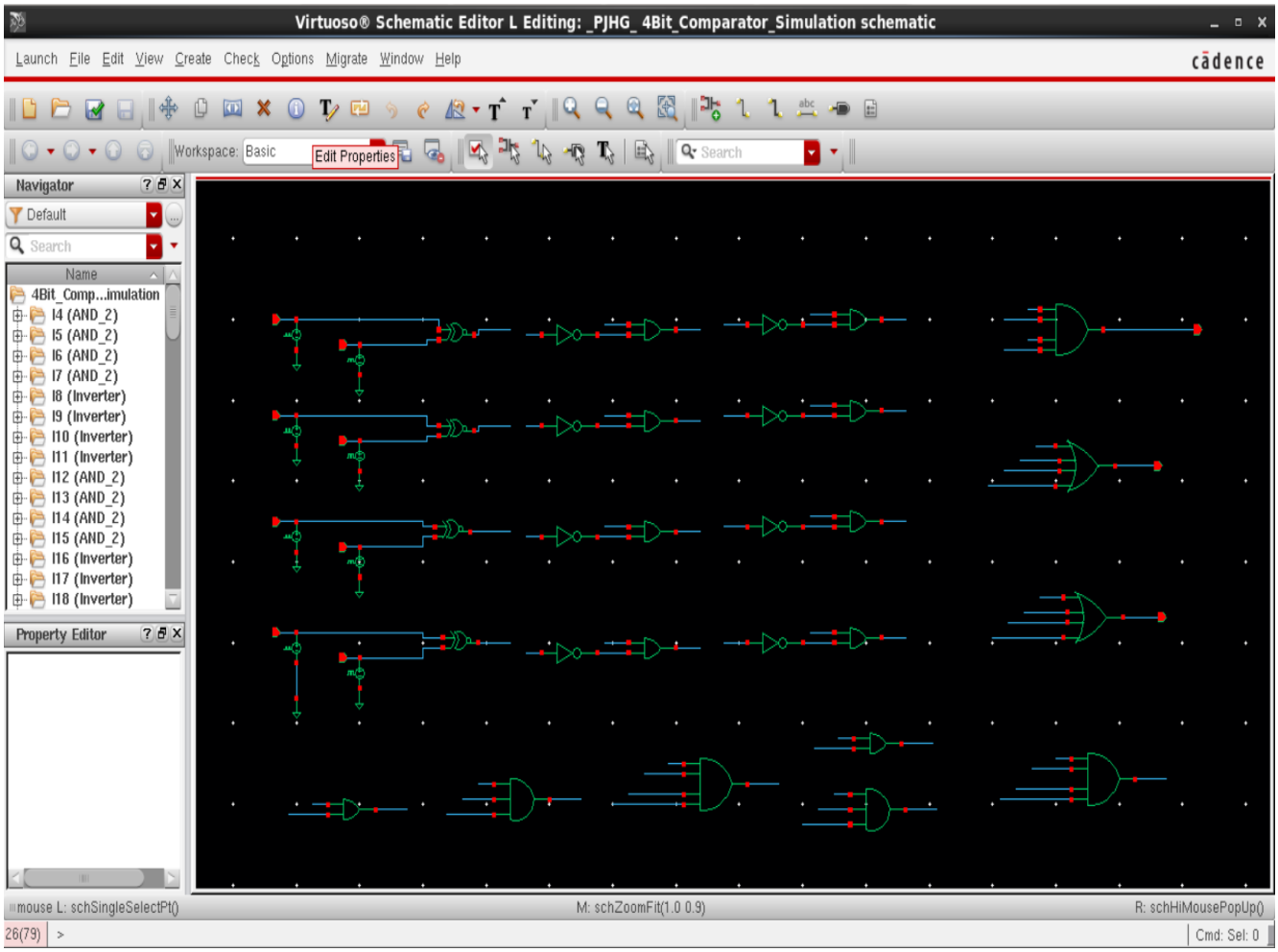


Figure 4.16: 4-bit Comparator Transistor-Level Schematic.

4.7.2 Symbol

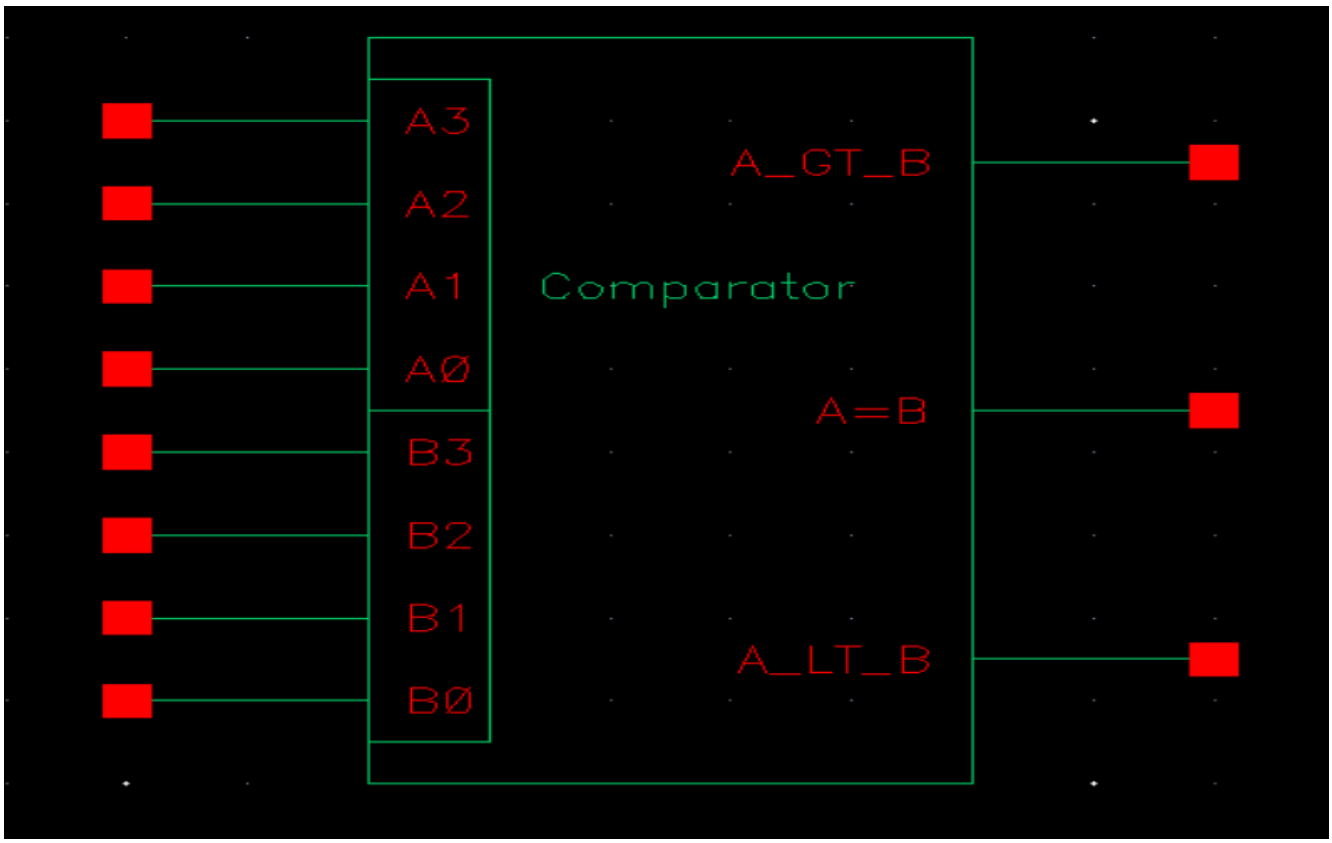


Figure 4.17: 4-bit Comparator Symbol.

4.7.3 Simulation

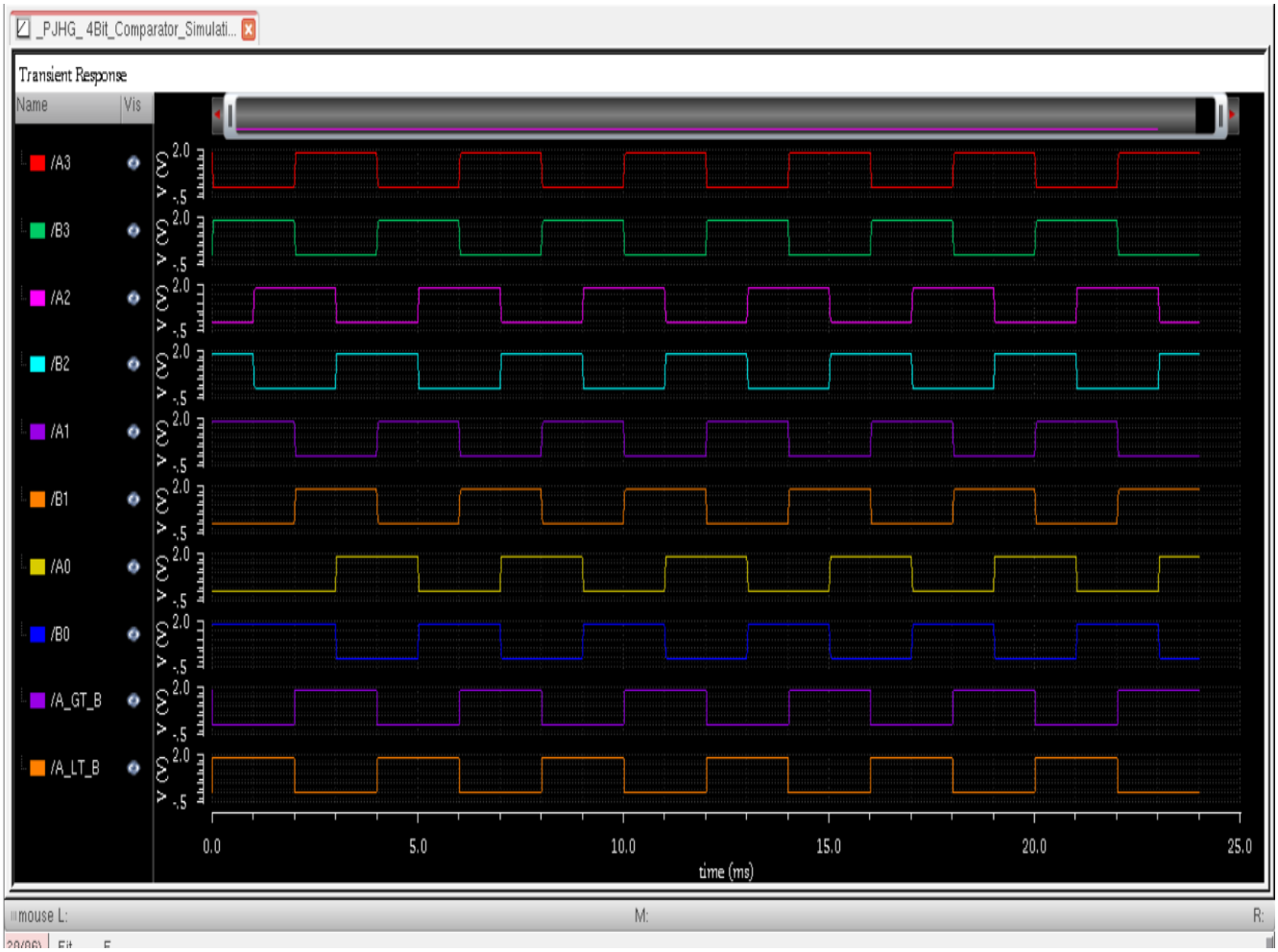


Figure 4.18: 4-bit Comparator Simulation Waveforms.

4.8 C^{2}MOS Flip-Flop Cell

4.8.1 Transistor-Level C^{2}MOS Flip-Flop Cell

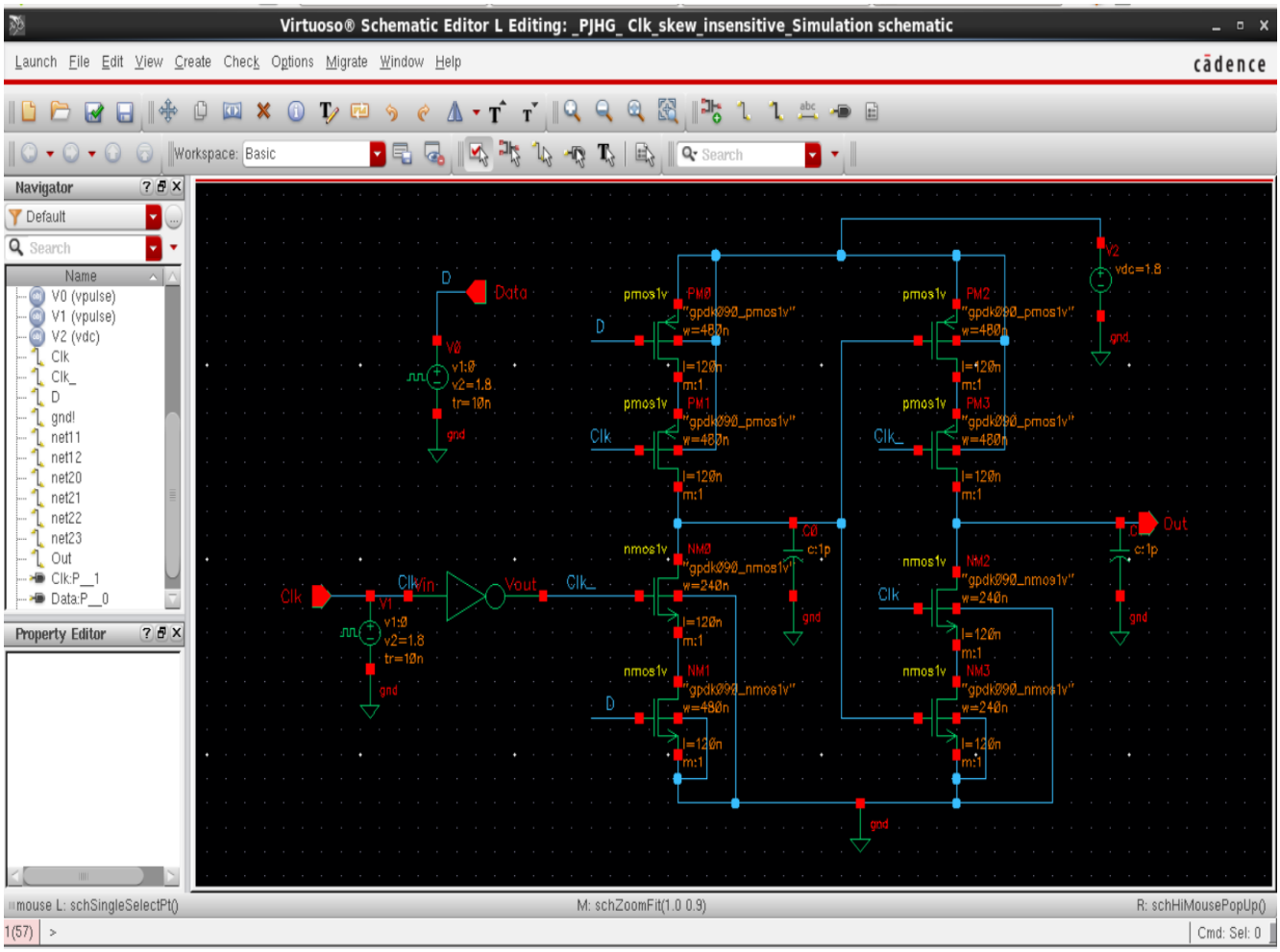


Figure 4.19: C^{2}MOS Flip-Flop Cell Schematic.

4.8.2 Symbol

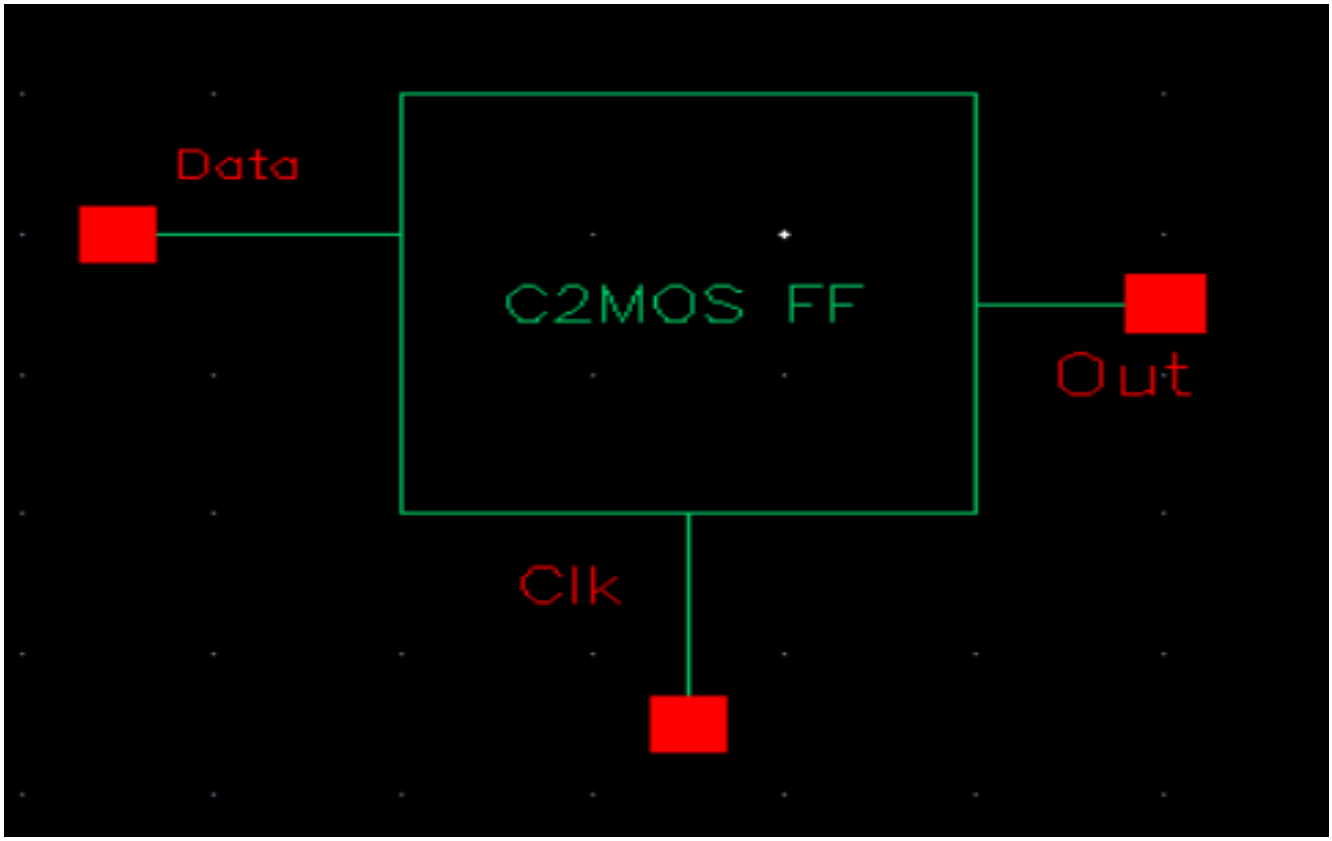


Figure 4.20: C^2MOS Flip-Flop Symbol.

4.8.3 Simulation



Figure 4.21: C^{2}MOS Flip-Flop Waveforms.

4.9 Johnson Counter

4.9.1 Transistor-Level Johnson Counter Cell

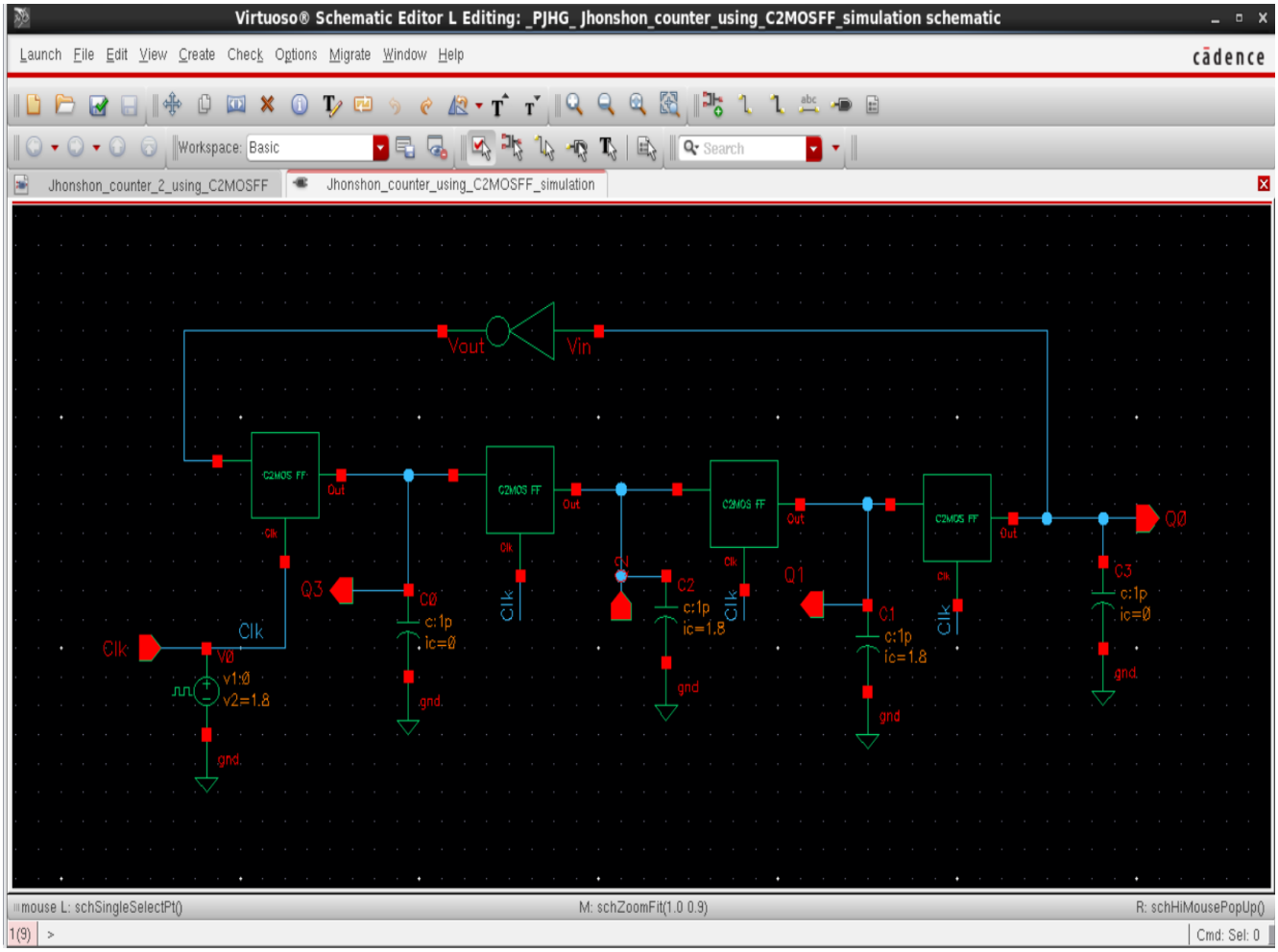


Figure 4.22: Johnson Counter 1 Schematic.

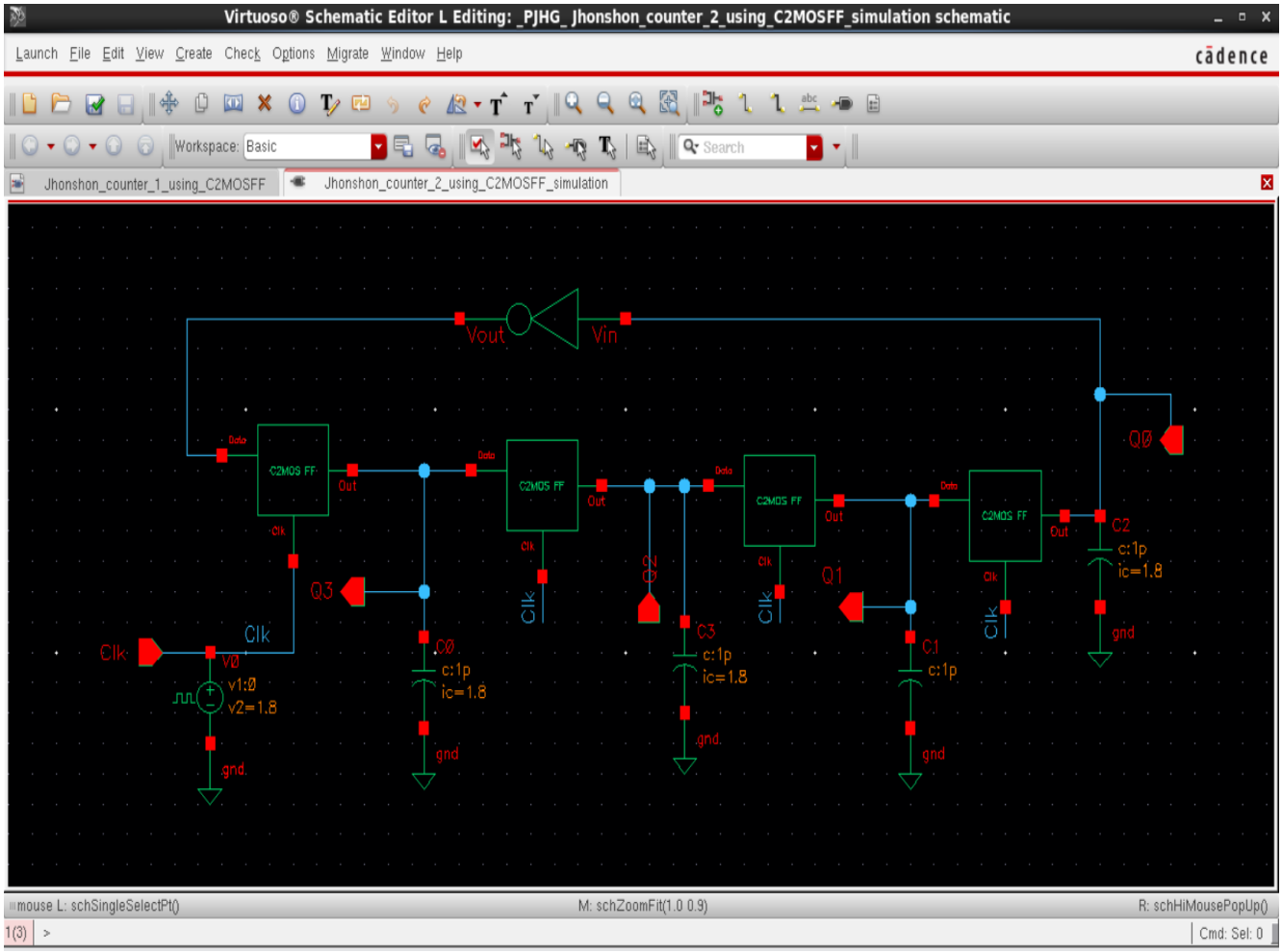


Figure 4.23: Johnson Counter 2 Schematic.

4.9.2 Symbol

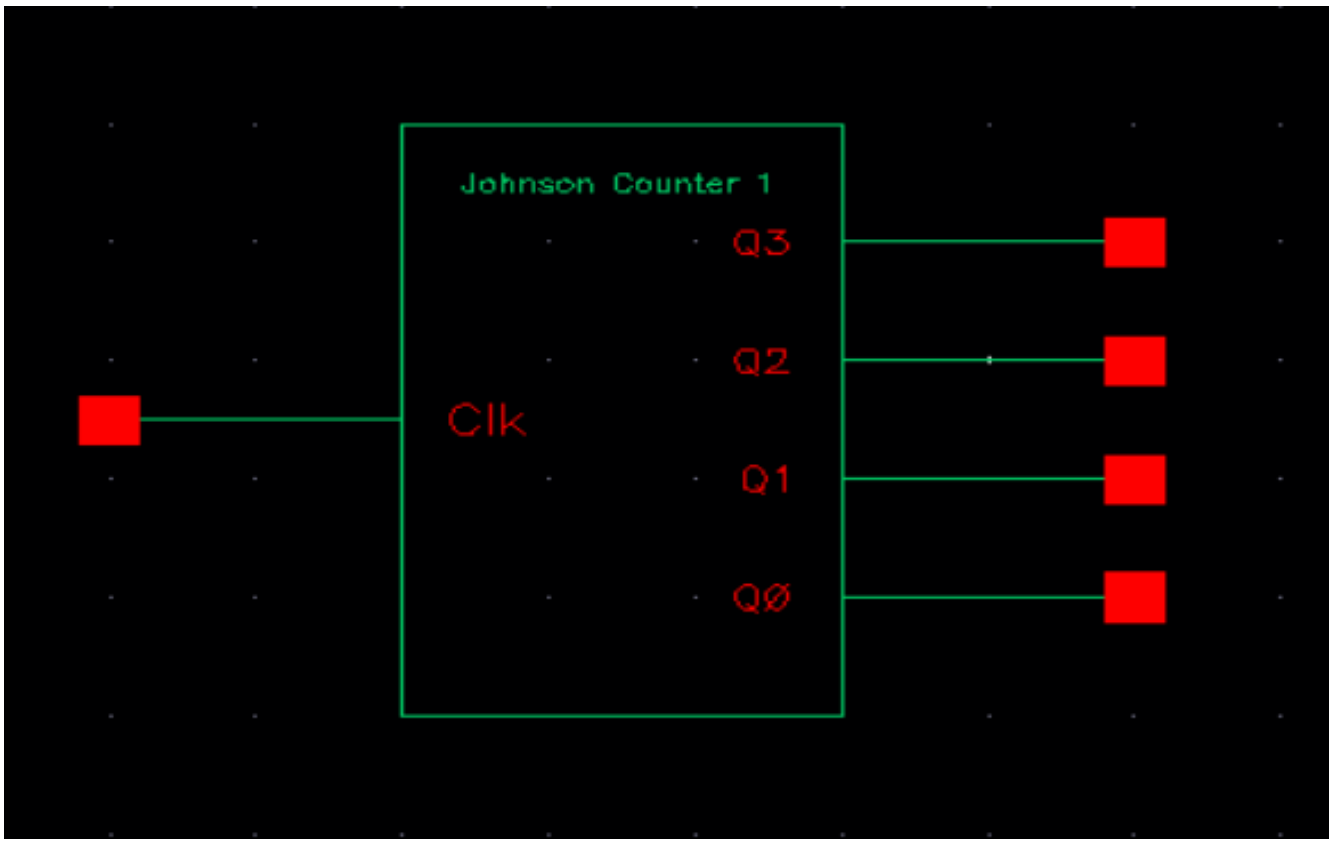


Figure 4.24: Johnson Counter 1 Symbol.

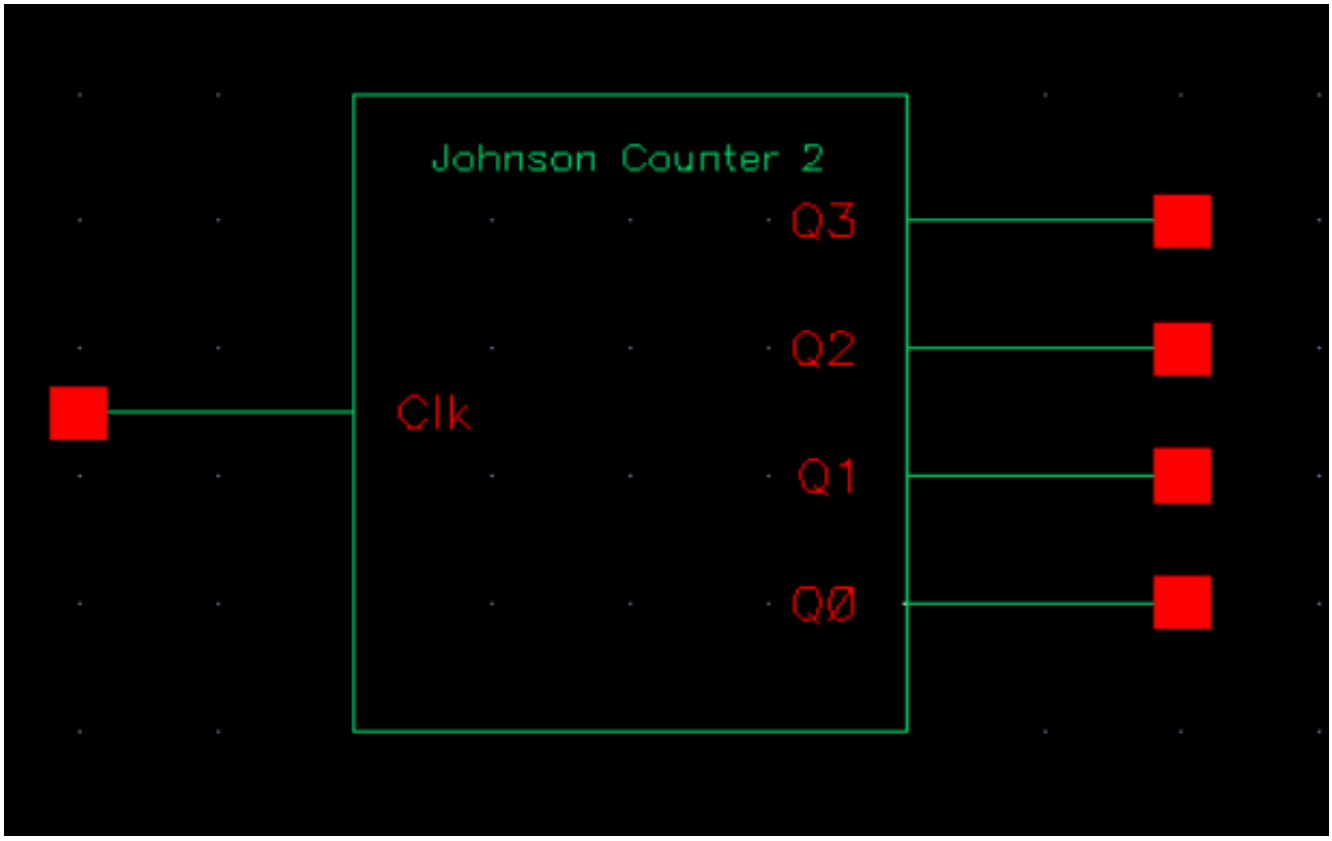


Figure 4.25: Johnson Counter 2 Symbol.

4.9.3 Simulation

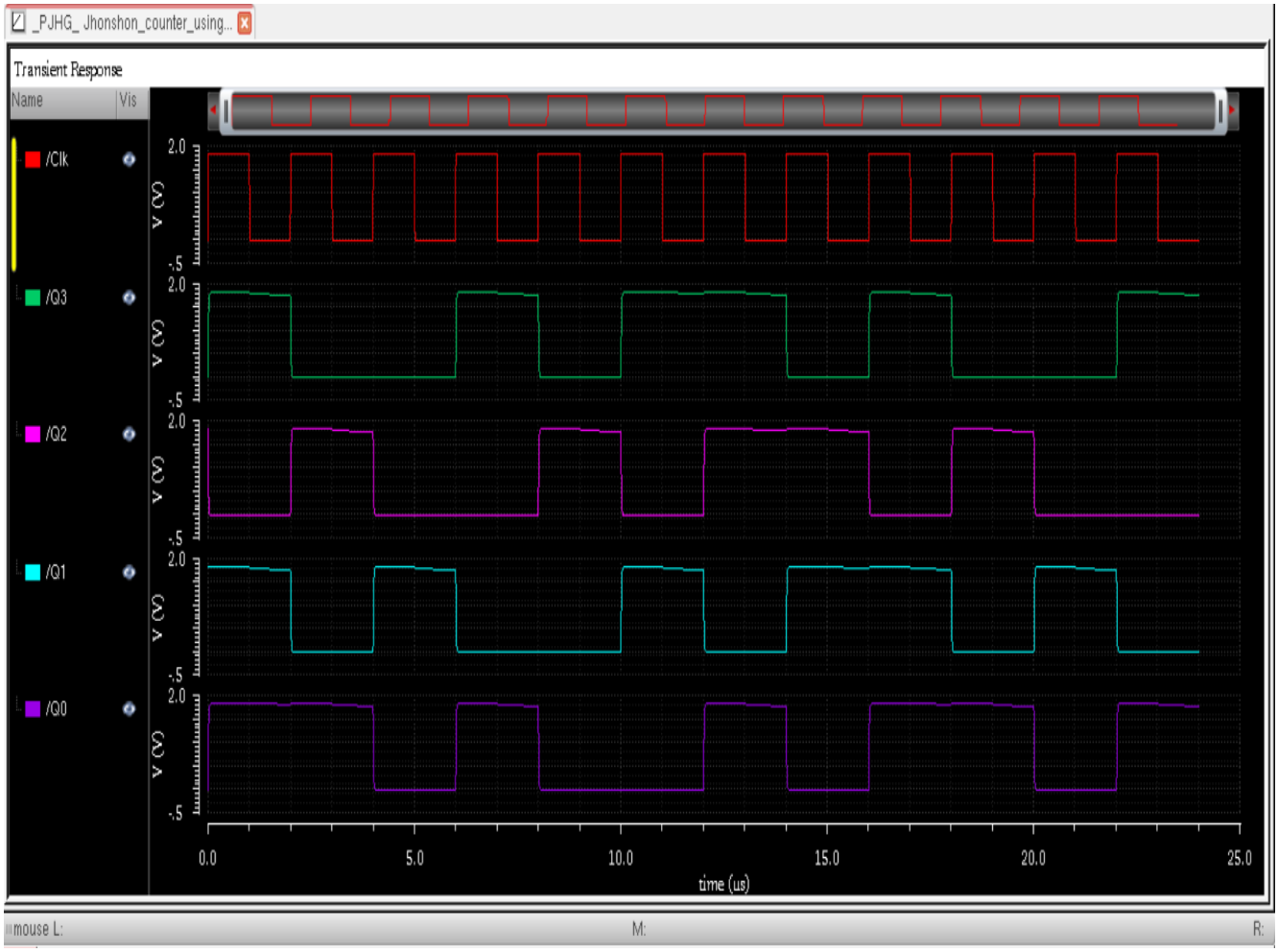


Figure 4.26: Johnson Counter 1 Waveforms.

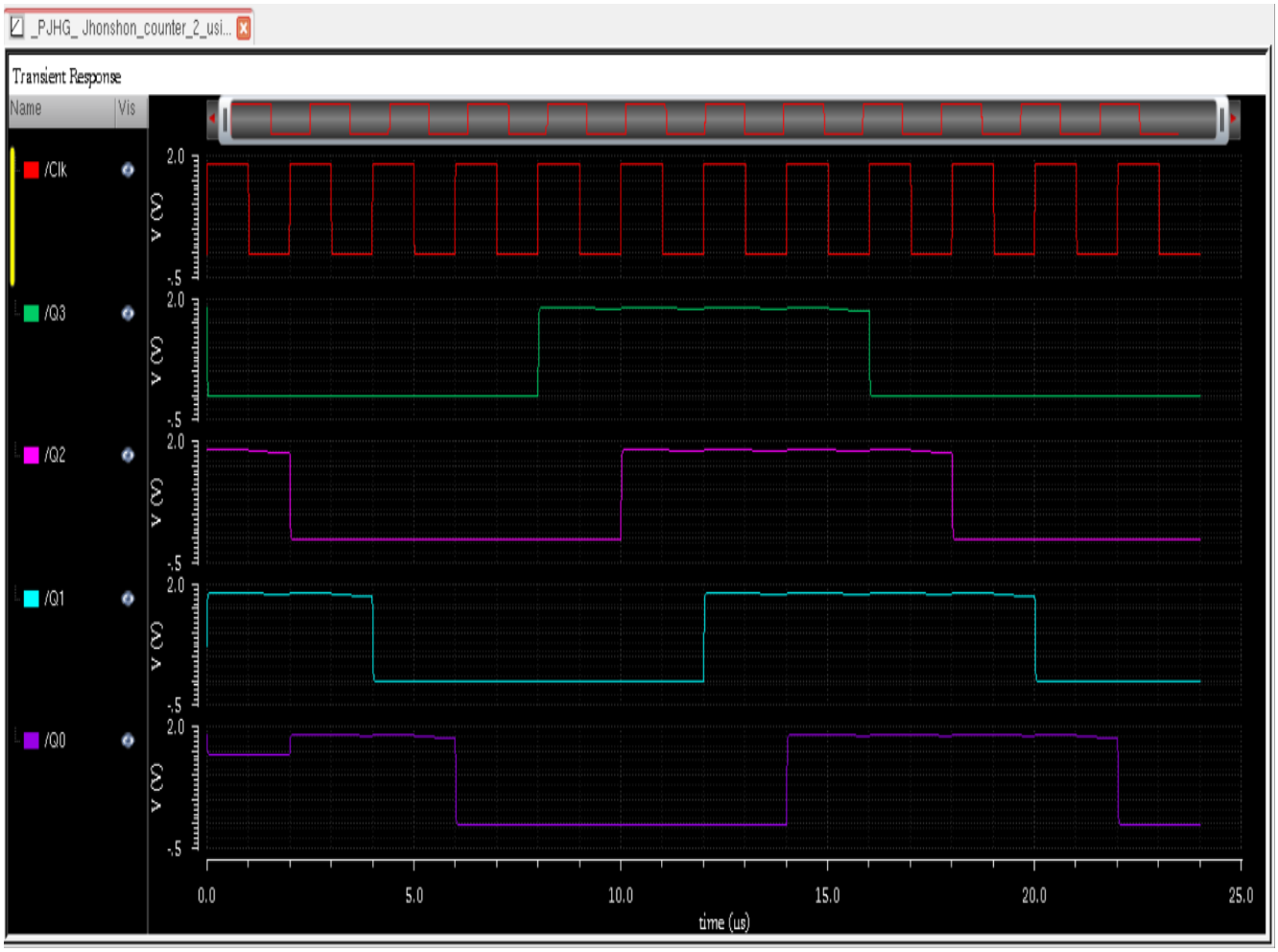


Figure 4.27: Johnson Counter 2 Waveforms.

4.10 6T SRAM Cell

4.10.1 Transistor-Level Memory Cell

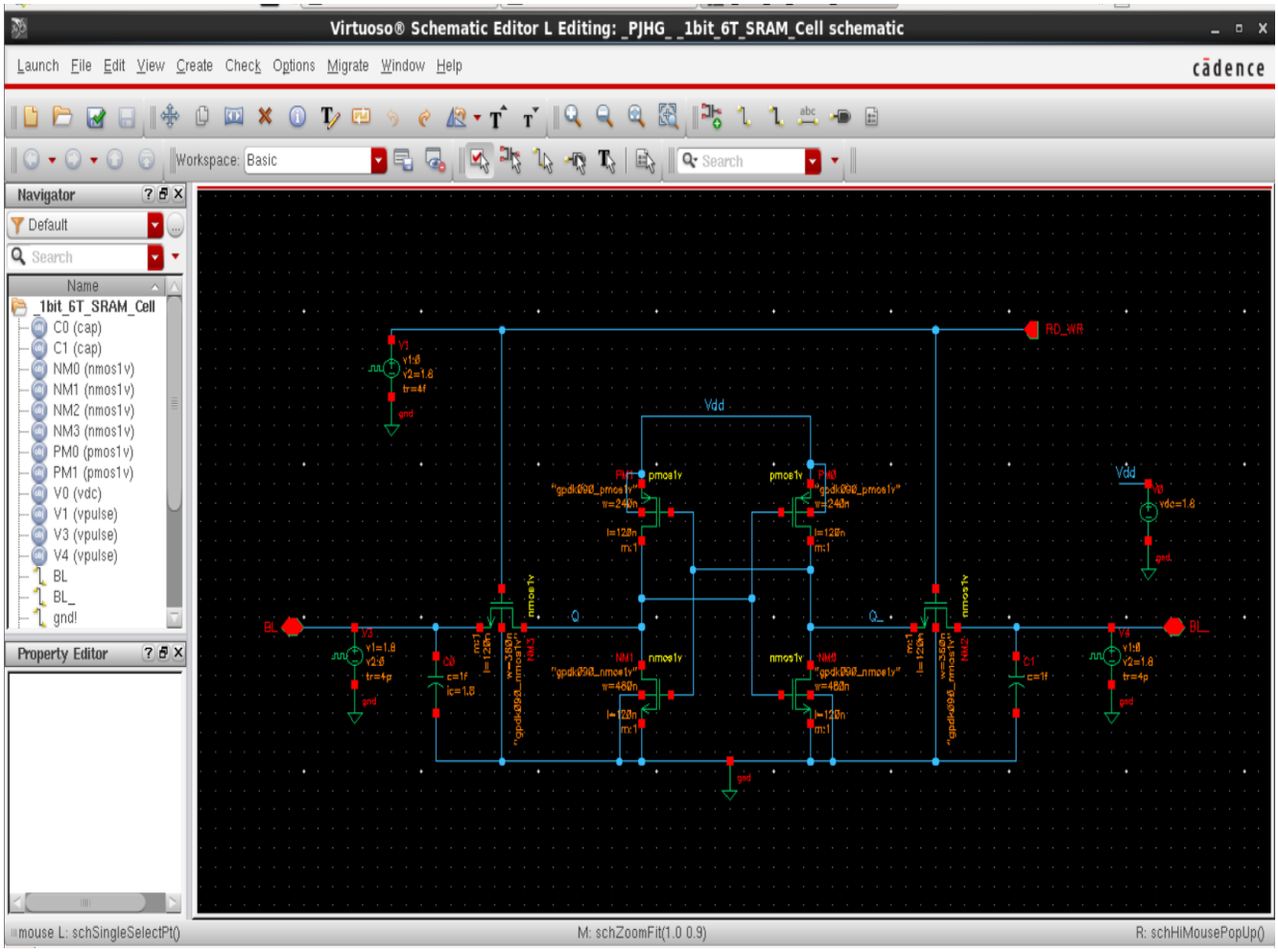


Figure 4.28: 6T SRAM Cell Schematic.

4.11 Sense Amplifier - OTA

4.11.1 Schematic

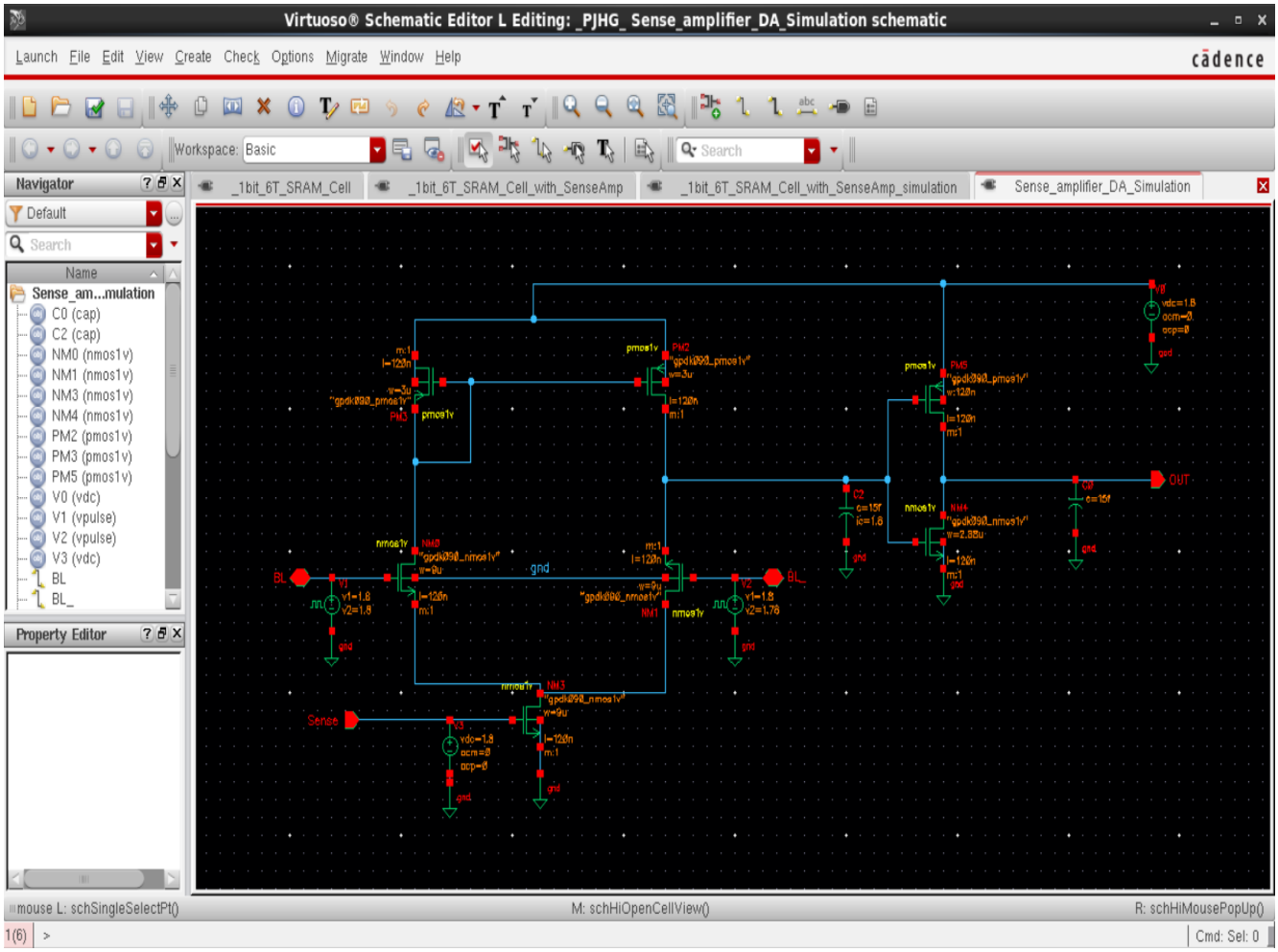


Figure 4.29: Sense Amplifier Schematic.

4.11.2 Symbol

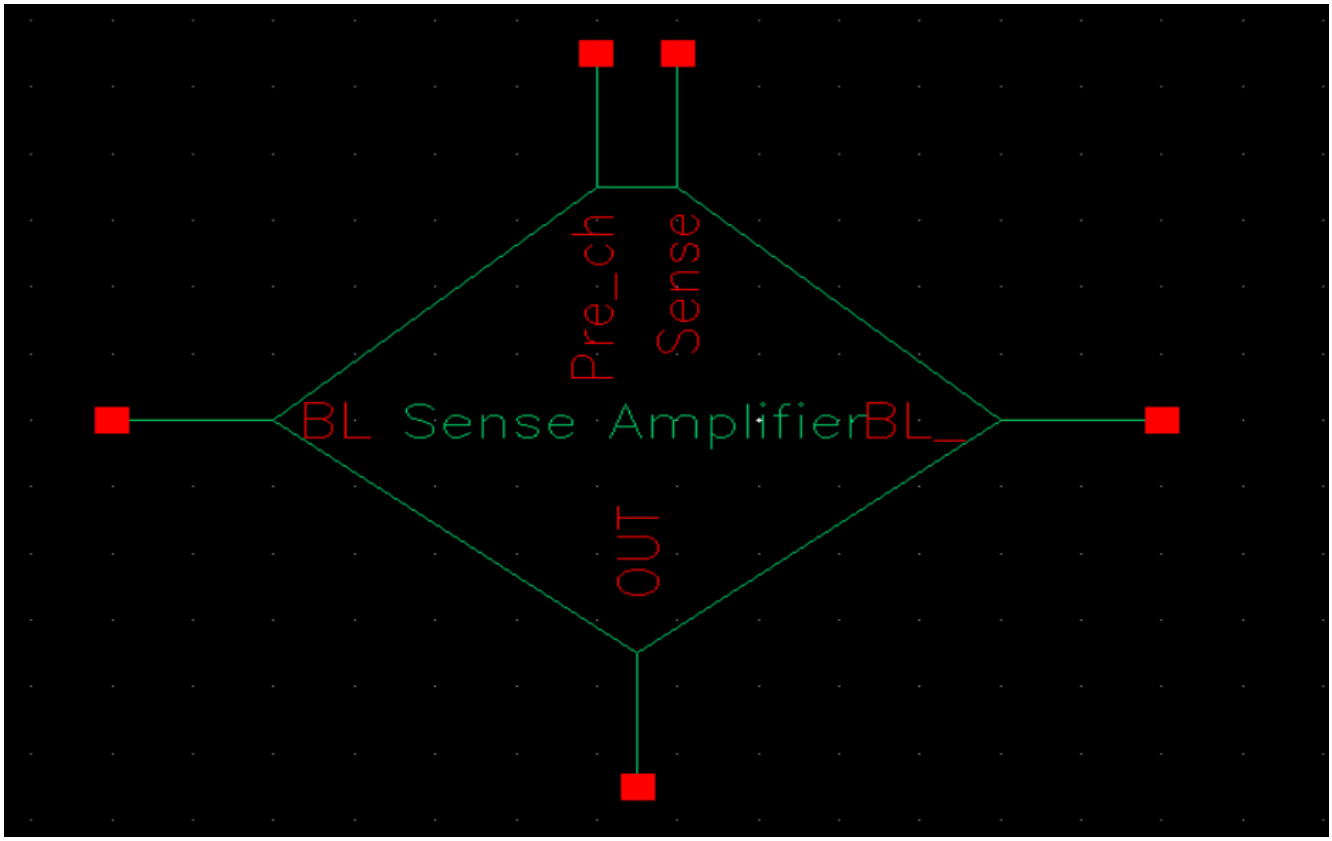


Figure 4.30: Sense Amplifier Symbol.

4.11.3 Simulation

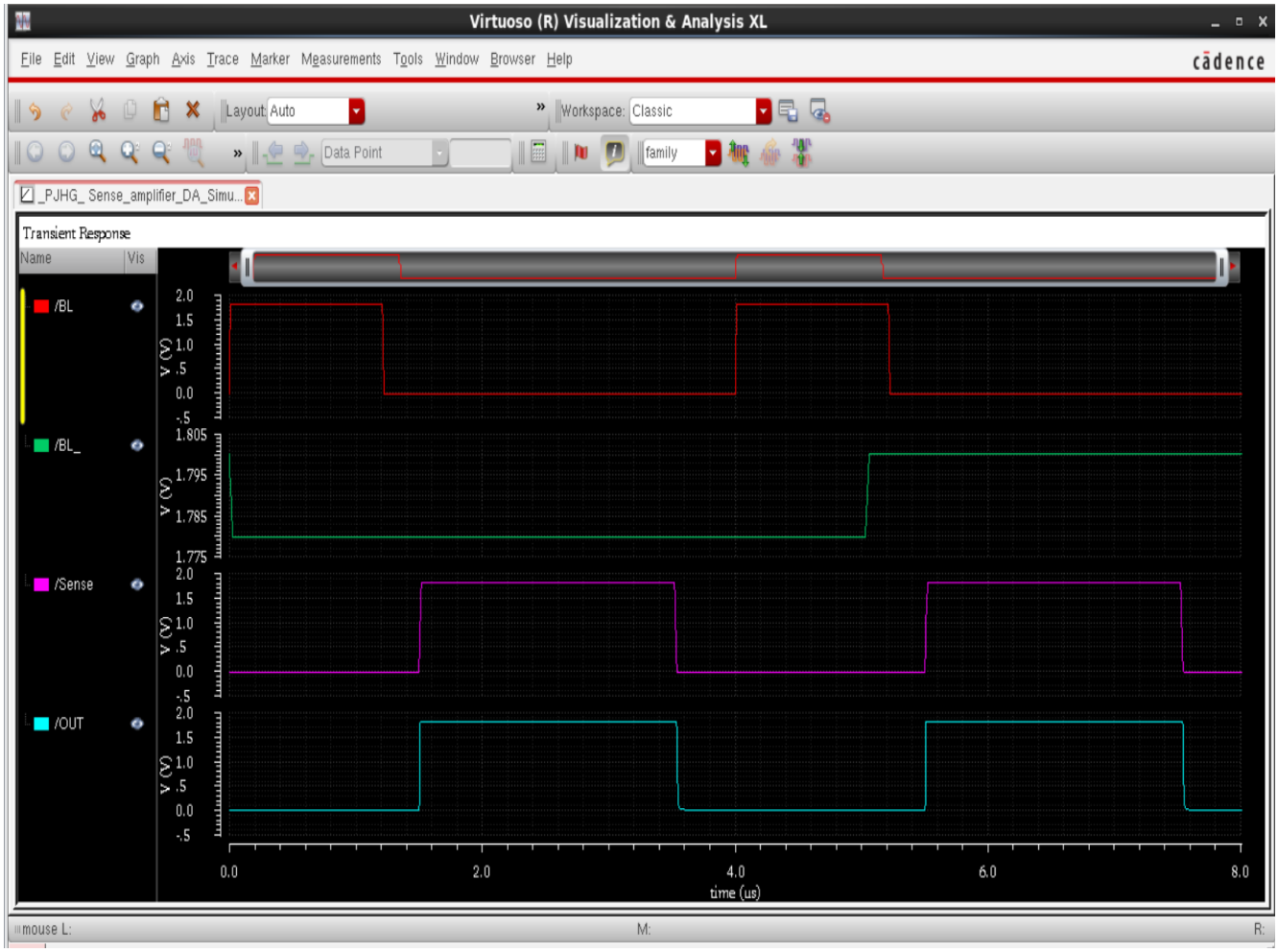




Figure 4.31: Sense Amplifier Simulation Waveforms.

4.12 Integrated 6T SRAM + Sense Amplifier Block

4.12.1 Combined Schematic

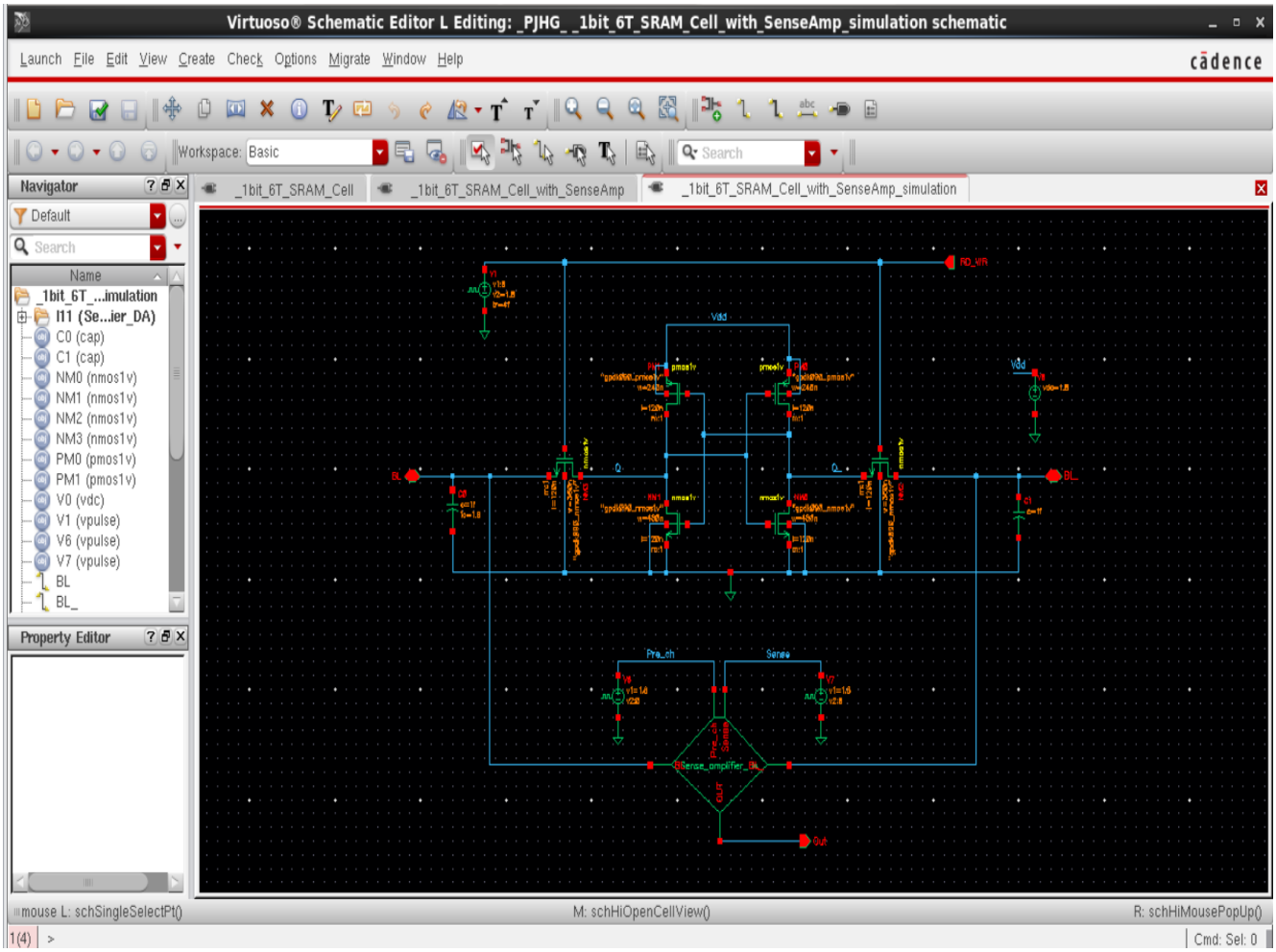


Figure 4.32: Integrated 6T SRAM + Sense Amplifier Schematic.

4.12.2 Simulation



Figure 4.33: Integrated 6T SRAM + Sense Amp Simulation Waveforms.

4.13 Stored Passwords - A and B

4.13.1 Passwords A Schematic

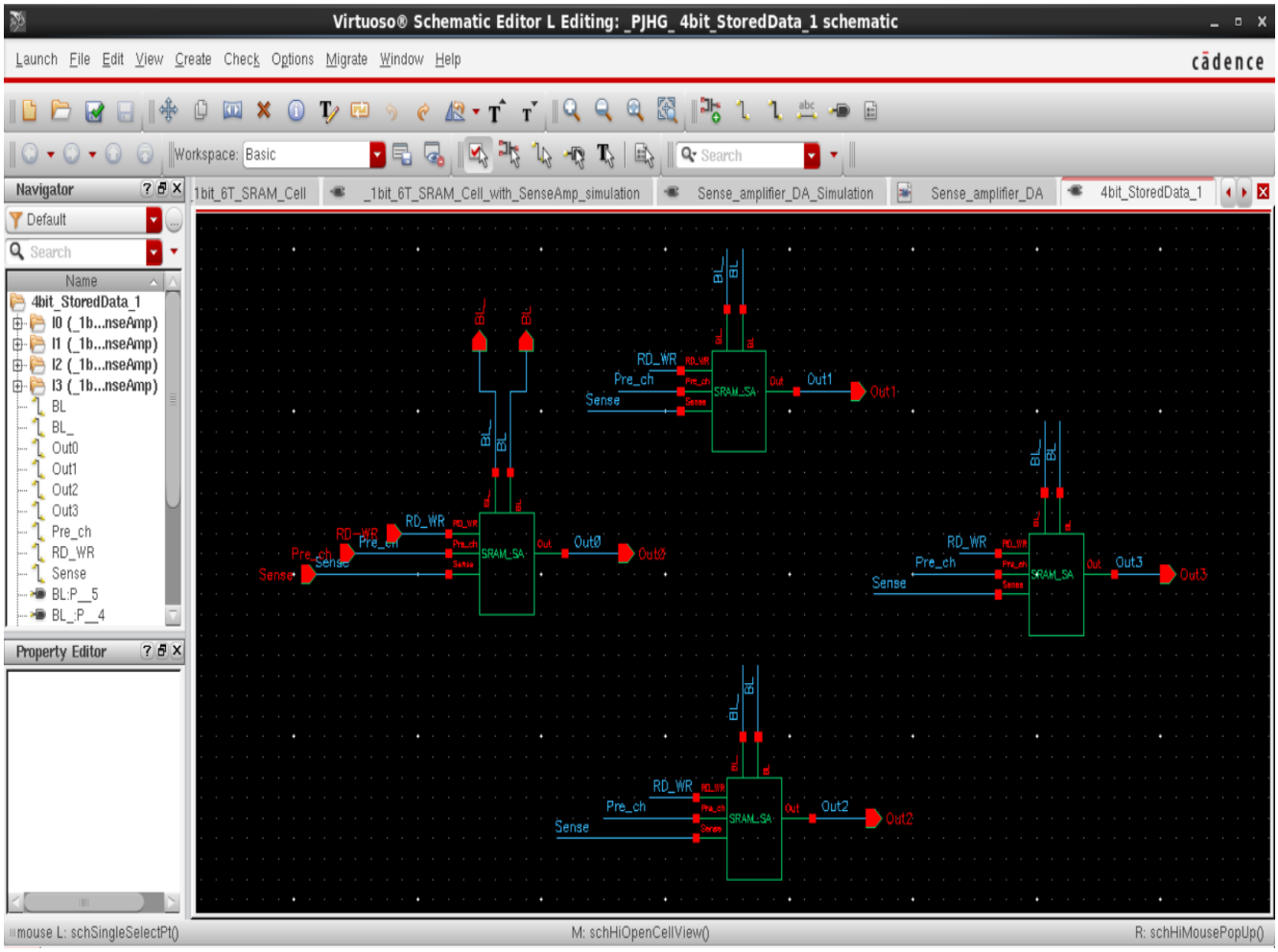


Figure 4.34: Schematic of 4-bit Password-A Stored in Memory

4.13.2 Passwords A Symbol

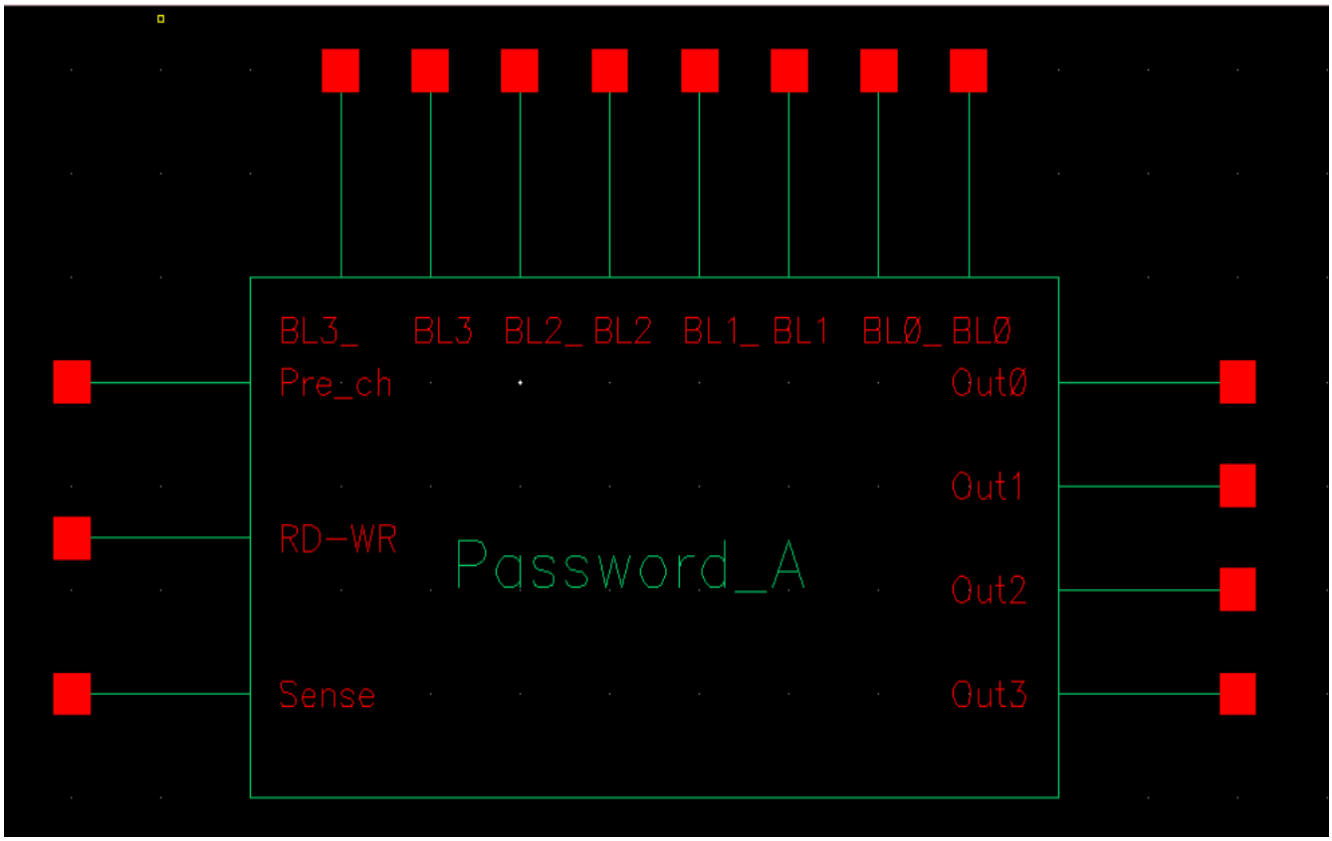


Figure 4.35: Symbol of 4-bit Password-A Stored in Memory.

4.13.3 Passwords B Schematic

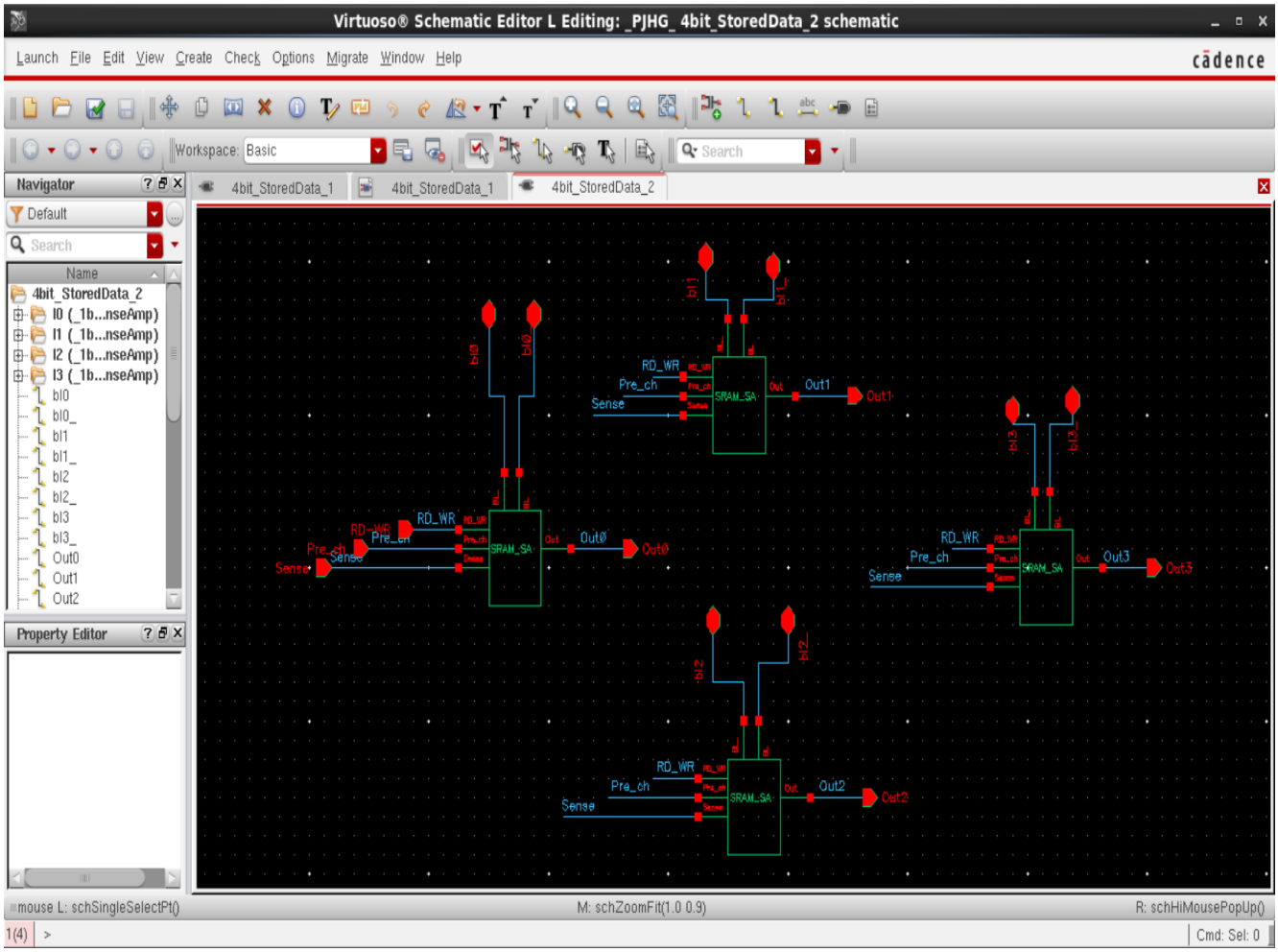


Figure 4.36: Schematic of 4-bit Password-B Stored in Memory

4.13.4 Passwords B Symbol

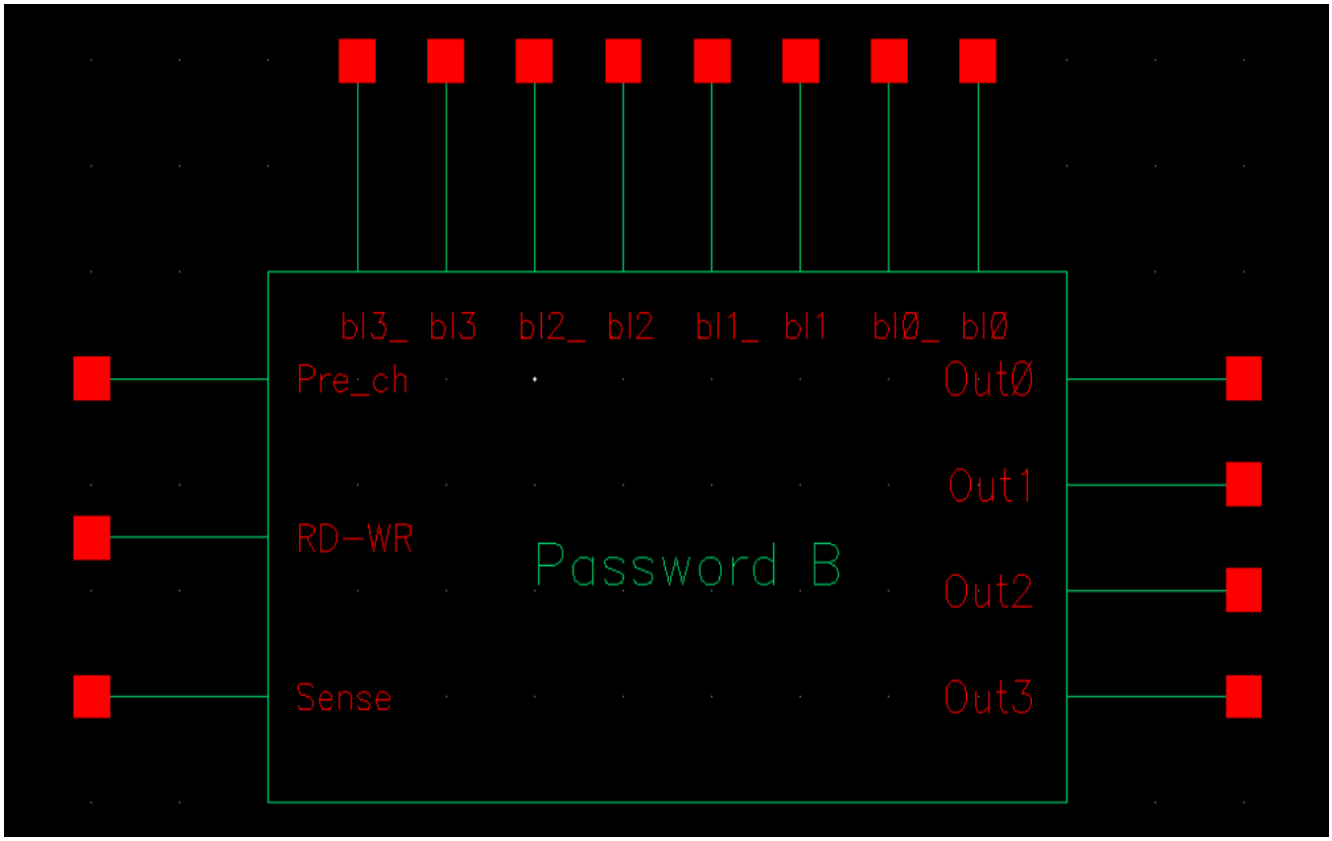


Figure 4.37: Symbol of 4-bit Password-B Stored in Memory.

Chapter 5

Complete Schematic

The complete transistor-level schematic integrates the following:

- Two 4-bit memory blocks
- Two 4-bit comparator blocks
- Final AND gate
- Two Johnson Counters Representing the User Inputs

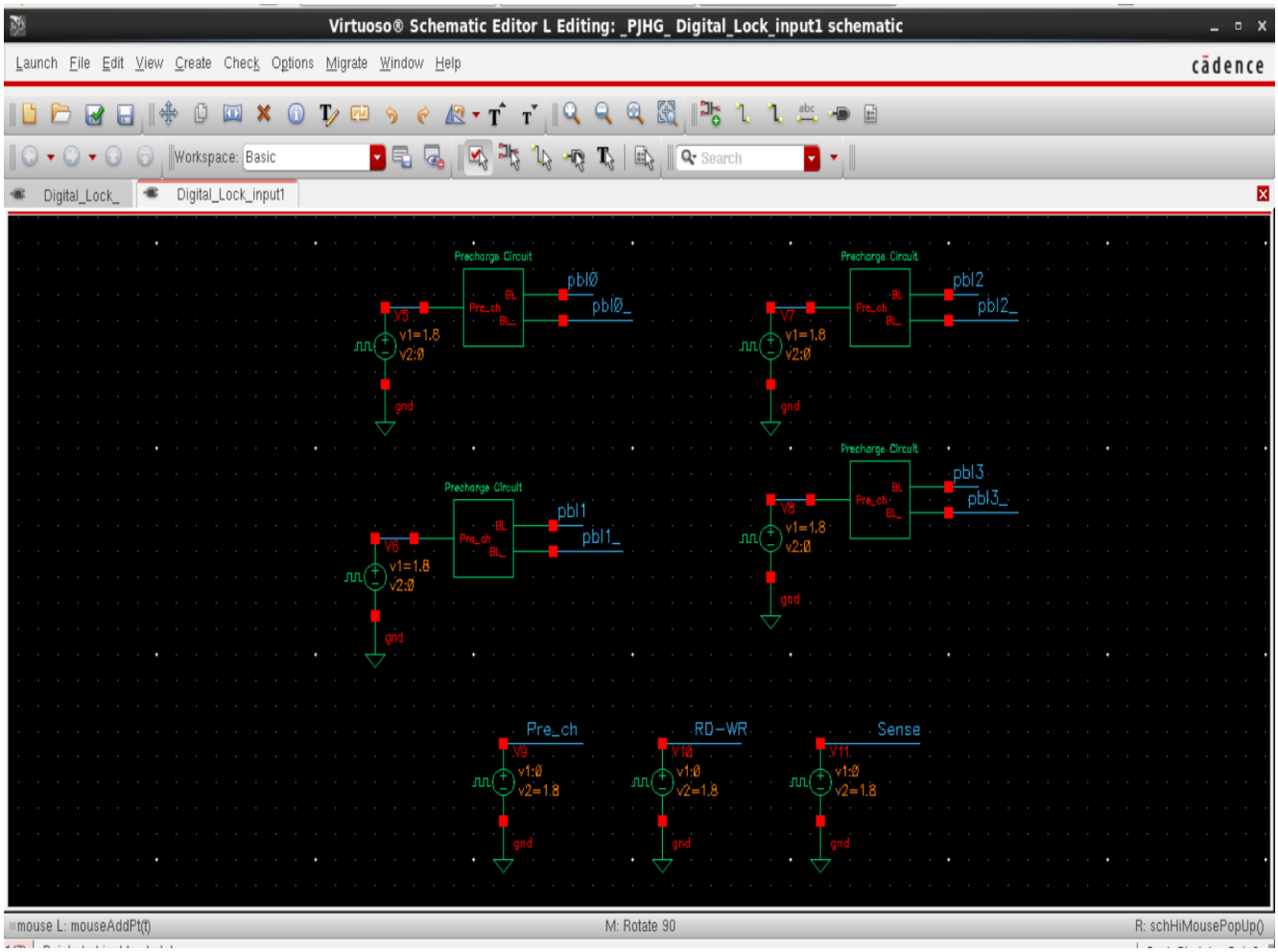


Figure 5.1: Input Block.

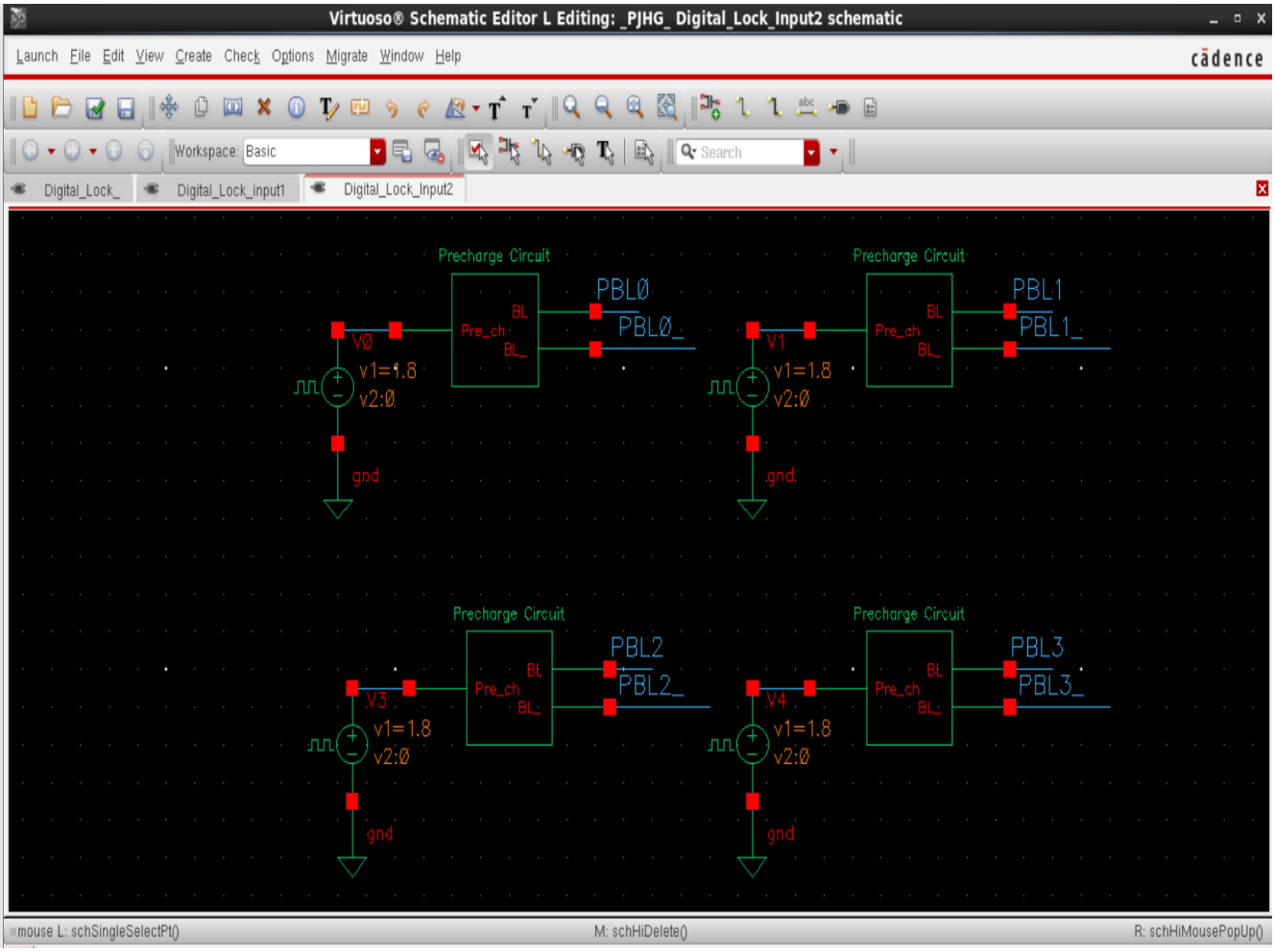


Figure 5.2: Input Block.

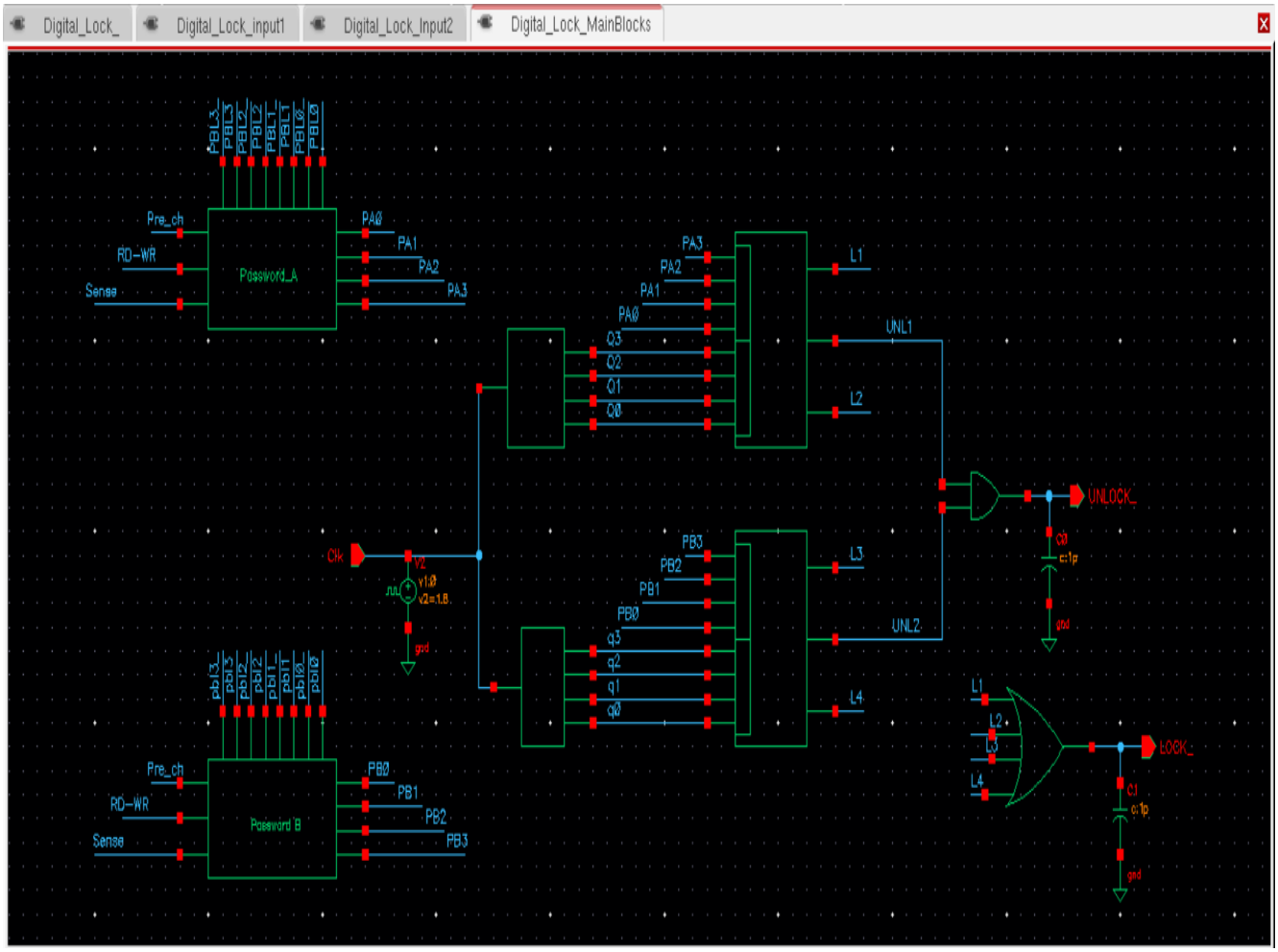


Figure 5.3: Main Blocks.

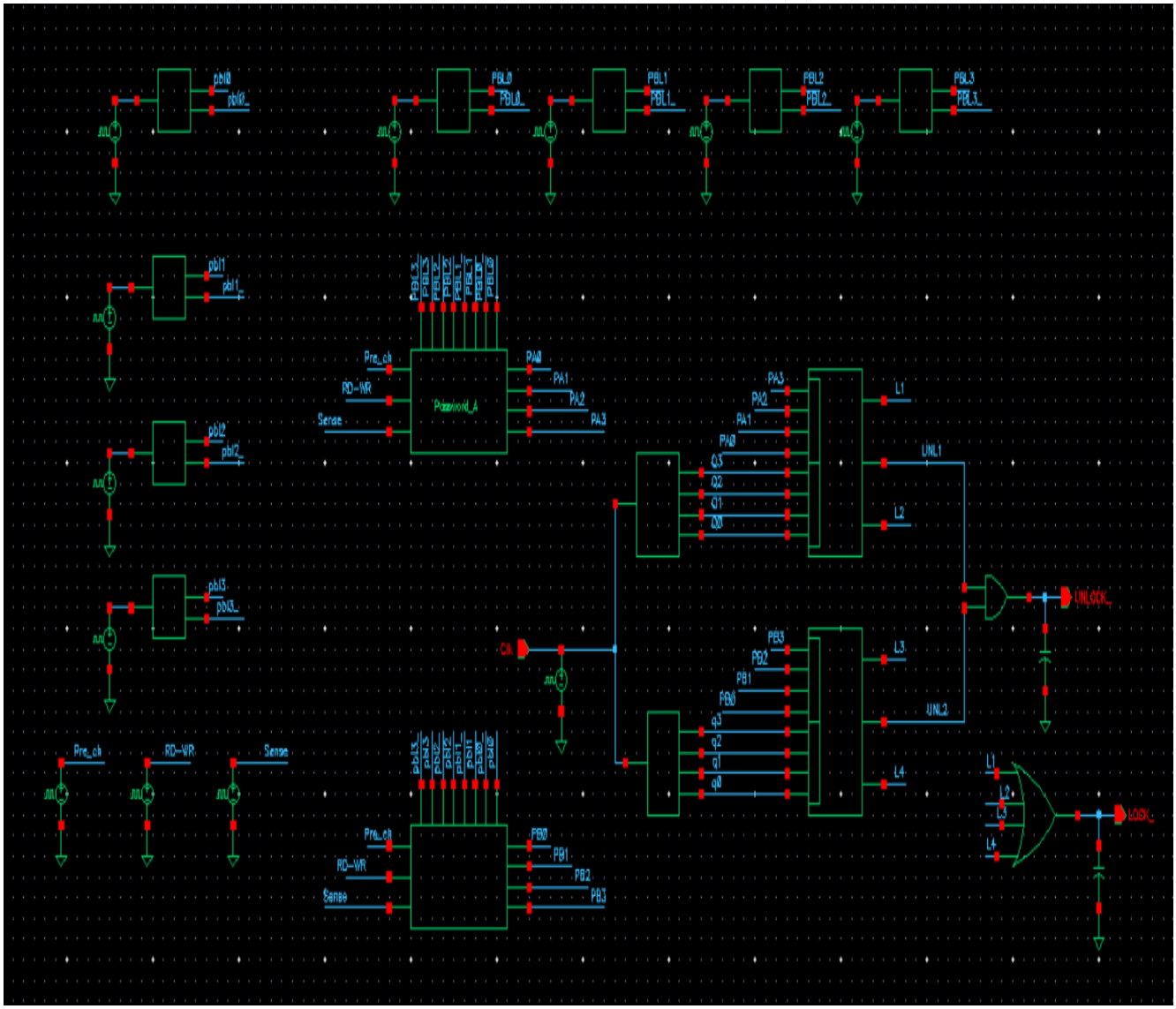


Figure 5.4: Complete integrated schematic.

Chapter 6

Simulation and Results

6.1 Simulation Setup

- **Supply voltage:** 1.8 V
- **Simulator:** Cadence Spectre (ADE L/XL)
- **Process Technology:** Corresponding PDK CMOS node
- **Supply Voltage** (V_{DD}): 1.8 V as per PDK
- **Temperature:** 27°C (typical corner)
- **Analysis Type:** Transient (time-domain)
- **Counter clock signal:** 500 kHz.
- **Pulse Width of Clock :** 50 percent.
- The system stores two predefined 4-bit Passwords: 1100 and 1010.

6.2 Block Waveforms and Observations



Figure 6.1: Simulation waveforms

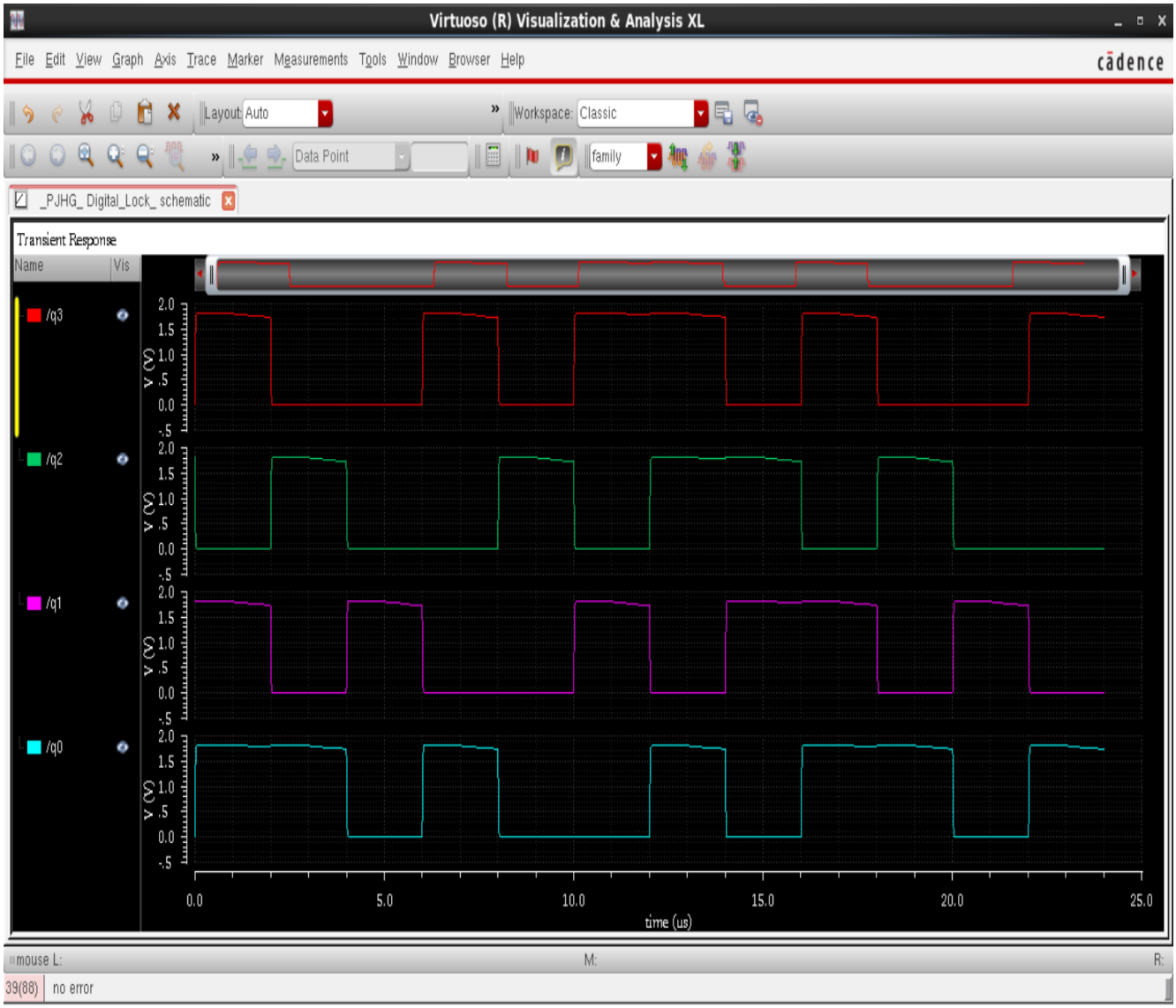


Figure 6.2: Simulation waveforms

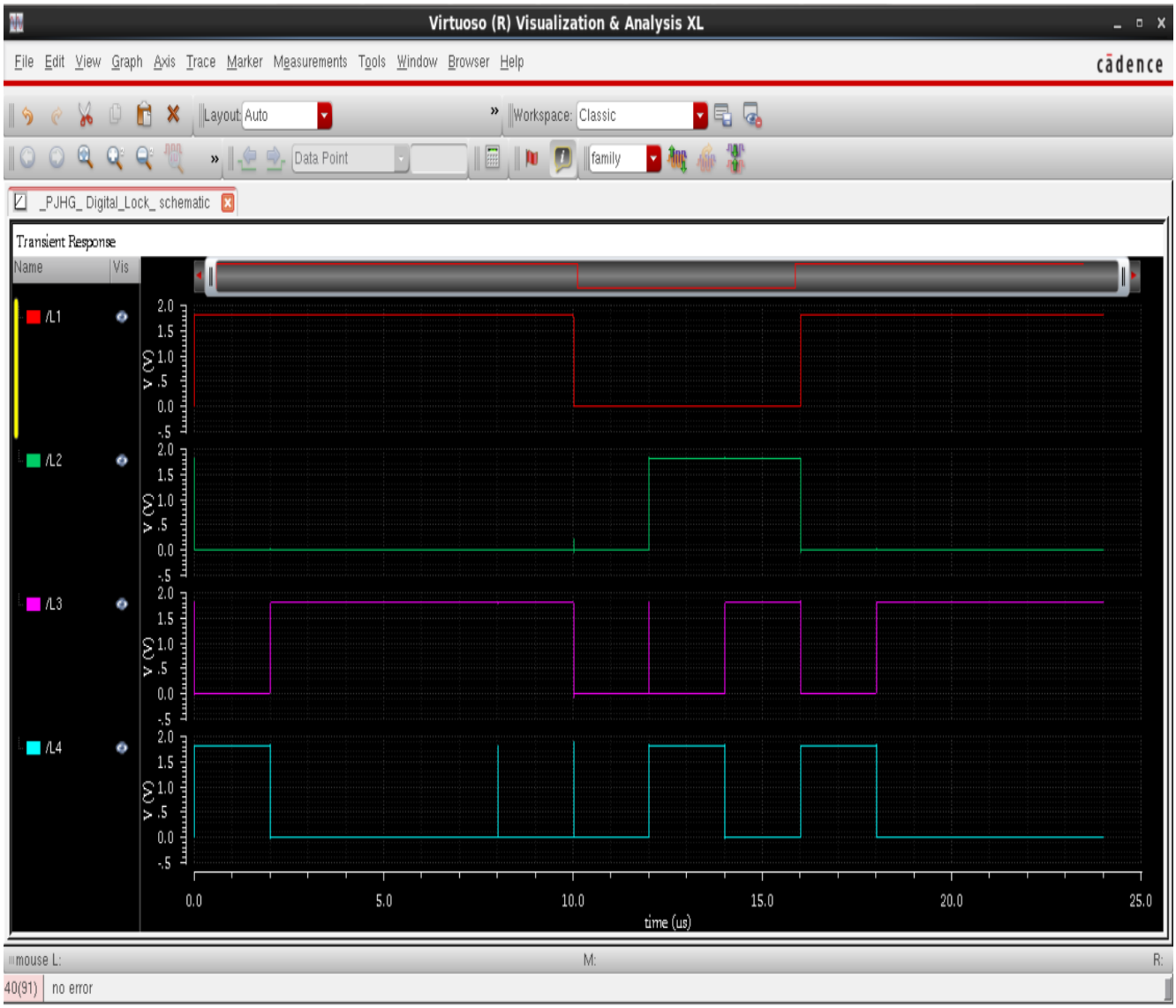


Figure 6.3: Simulation waveforms

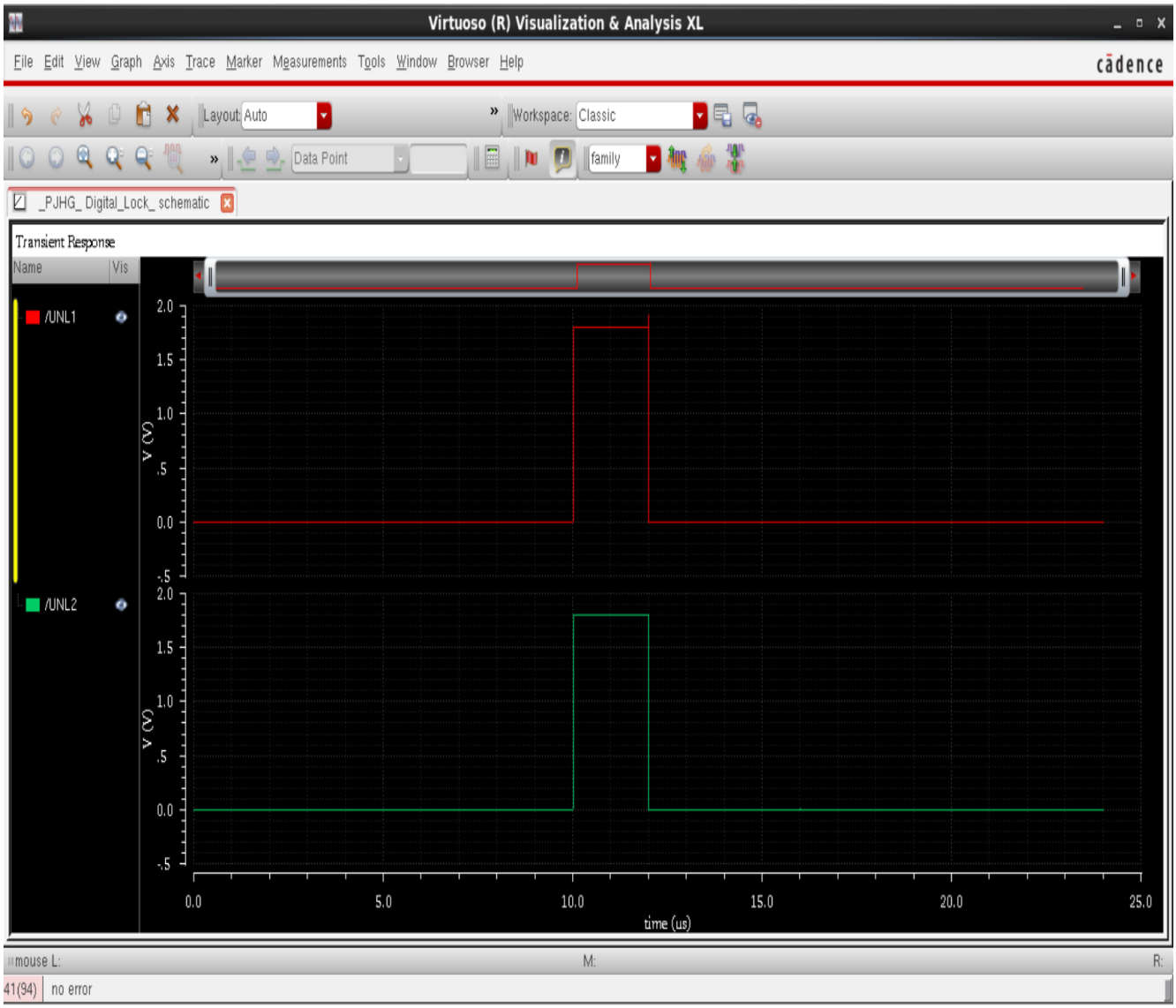


Figure 6.4: Simulation waveforms

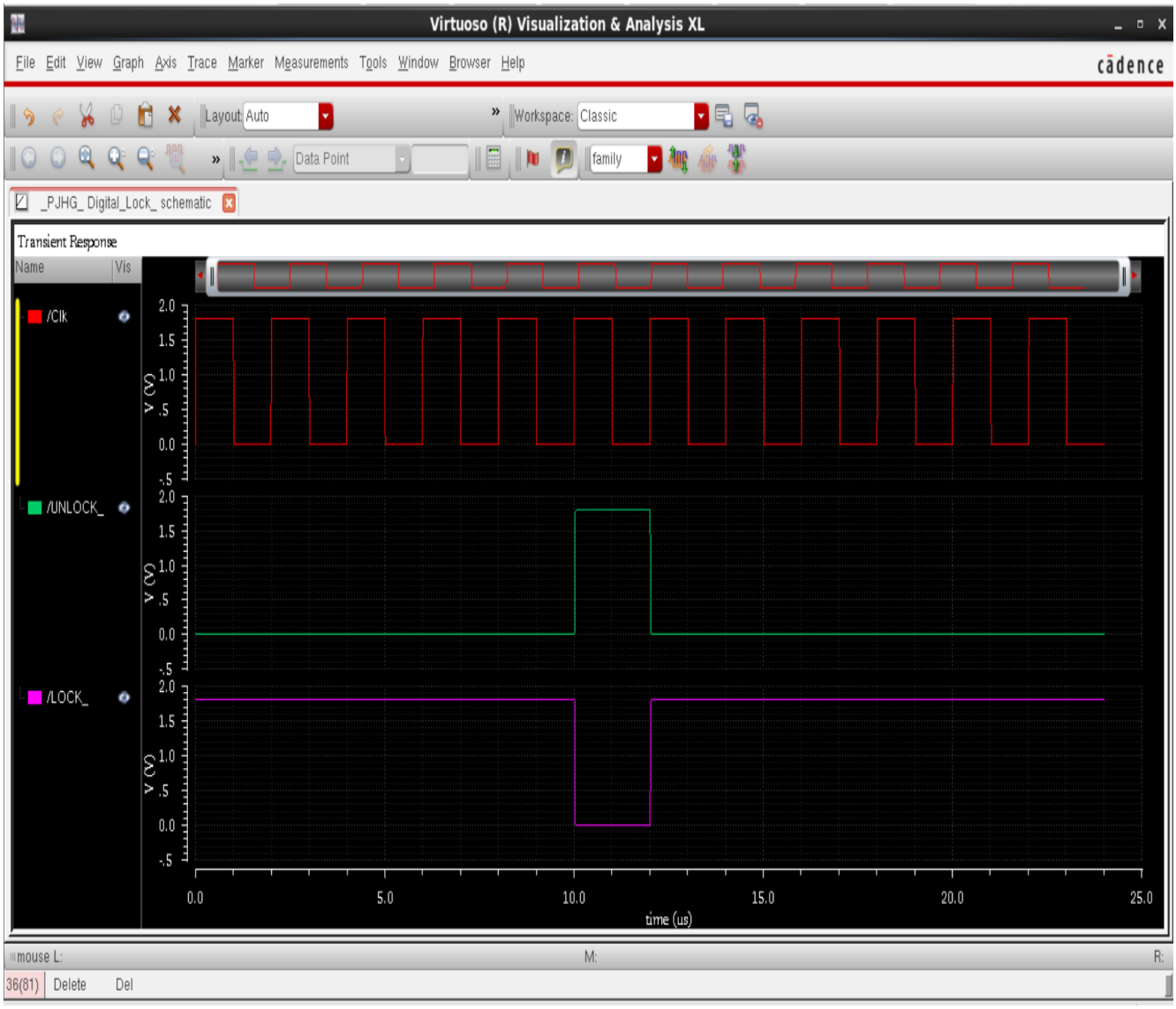


Figure 6.5: Simulation waveforms

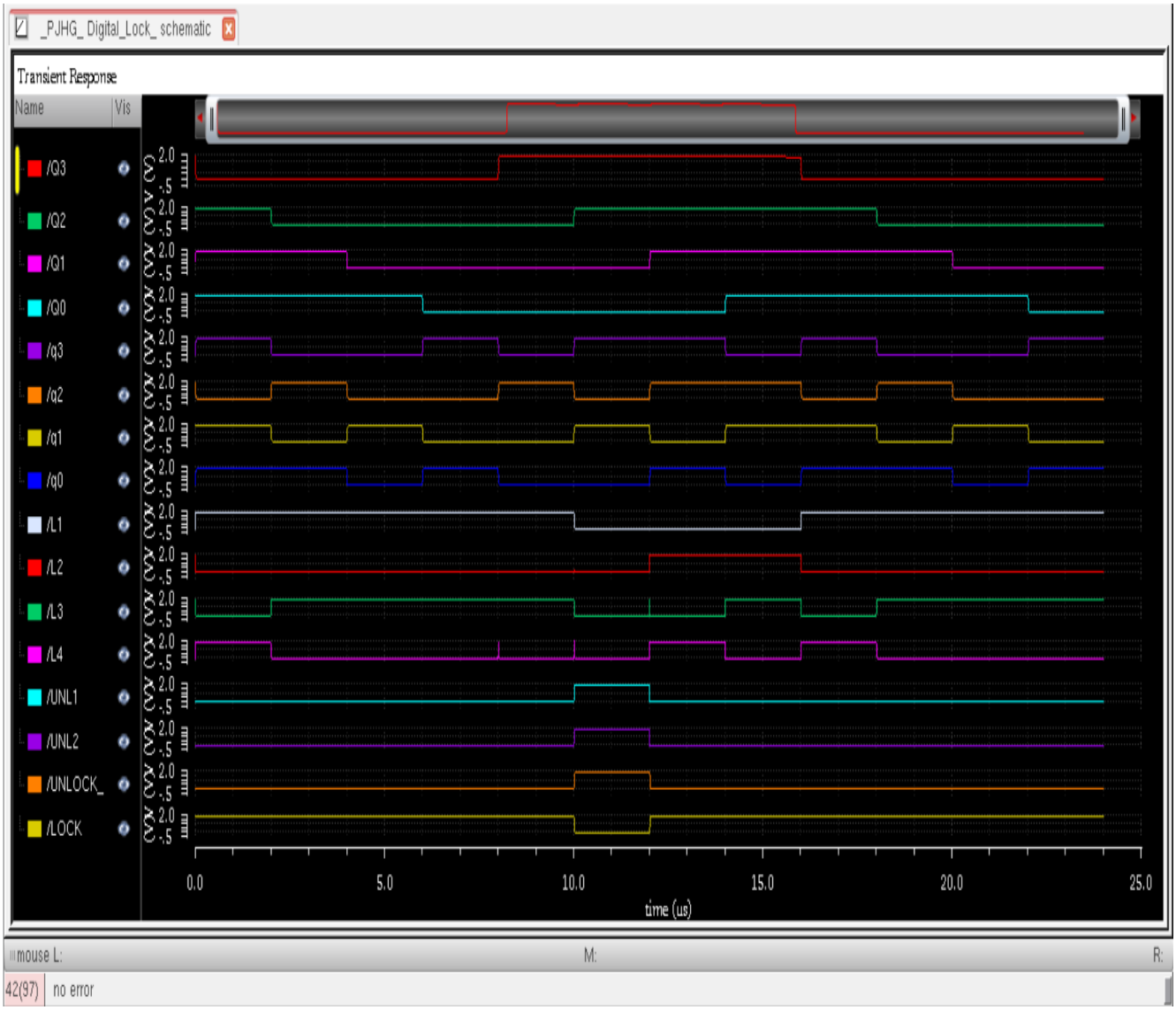


Figure 6.6: Simulation waveforms

Chapter 7

Conclusion

A 4-bit transistor-level Digital Lock System was successfully designed and simulated in Cadence Virtuoso. Two stored passwords against Johnson-counter-generated inputs were compared using transistor-level comparators, and the lock was opened only when both comparisons matched.

References

- Rabaey, Jan M., et al. *Digital Integrated Circuits*.
- Weste, Neil H. E., *CMOS VLSI Design*.
- Cadence Virtuoso and PDK documentation.